CAMPUSOS

INDIA-FIRST / K–12 / MULTI-TENANT SAAS

A calmer **school day.**

The complete product, engineering and experience blueprint.

One connected system for school leaders, teachers, families and the teams who keep everything running.

24
feature domains

23
screen concepts

144
requirements



PRODUCT REQUIREMENTS + UI DESIGN + DELIVERY PLAN

V1.0 • 10 NOVEMBER 2026

A complete plan, built to be used.

An implementation blueprint and visual design direction. All screen data and school names are fictional.

What this deliverable contains

A prioritized feature catalogue with stable IDs; business rules and acceptance checks; SaaS operations; technical architecture; data and API contracts; and 23 designed screens.

How to read the priorities

P0 — launch: a usable, secure school product.
P1 — expansion: deeper operations and automation.
P2 — optional: enterprise and specialist modules.
Priorities indicate sequence, not missing scope.

Working assumptions

India-first K–12; independent schools and school groups; responsive web first; English at launch with localization foundations; INR and Asia/Kolkata defaults; configurable school calendars.

What still needs validation

Confirm pilot school workflows, board-specific templates, fee/refund policies, legal applicability, retention periods, scale and provider contracts. Mockups are design targets; they are not a functioning application.

Navigate the blueprint

PART	CONTENTS
01 / Product · 3–8	Scope, roles, permissions, navigation, journeys and commercial model
02 / Requirements · 9–32	24 modules • 144 requirements with rules and acceptance checks
03 / Engineering · 33–44	Architecture, data, APIs, security, delivery and operational readiness
04 / Experience · 45–71	Design system, 23 screen concepts, state patterns and sources

Built around the whole school.

A single operational record with purposeful experiences for each role.

Customer & job to be done

Sell to school owners and groups; support principals, administrators and staff as daily users. Families receive a simple portal for learning, communication and payments.

Product boundaries

Support the full student lifecycle and a commercial SaaS operating model. Launch with a fee subledger; grow into deeper finance, payroll and operations after validated core workflows.

Success measures • proposed pilot targets

OUTCOME	MEASURE	INITIAL TARGET
Reliable start	Required setup checklist completed	100% before go-live
Daily adoption	Class registers finalized by school cutoff	≥ 95% on school days
Financial clarity	Provider captures with one matched receipt	100%; exceptions owned
Family access	Invited guardians who complete activation	≥ 80% within 30 days
Operational value	Monthly reconciliation time vs baseline	30% reduction in pilot
Trust	Cross-tenant exposure / duplicate postings	0 accepted defects

Targets are proposed acceptance hypotheses, not measured product results. Baseline and approve them with pilot schools.

The right workspace for each role.

Role templates reduce setup effort; scoped permissions remain the source of truth.

ROLE	PRIMARY JOB	DEFAULT SCOPE
School owner / group leader	School configuration, governance and campus comparison	Authorized tenant or school group
Principal / academic coordinator	Daily exceptions, approvals, curriculum and outcomes	Assigned campuses
Registrar / admissions officer	Applications, records, enrolment and certificates	Assigned campuses / admissions intake
Teacher / substitute	Attendance, lessons, homework and assigned marks	Assigned classes, subjects and dates
Finance officer / accountant	Fees, collection, reconciliation and close	Assigned finance scope; no counselling notes
HR / payroll approver	Employment, leave, salary and payroll approval	Restricted staff fields
Librarian / transport / office	Operate assigned campus services	Relevant service records only
Nurse / safeguarding lead	Care, sensitive cases and emergency escalation	Explicit care/case permissions
Guardian / student	Own family tasks / own learning and results	Verified relationship / own record
SaaS operator / support	Subscriptions, service health and approved assistance	Metadata; temporary approved data access

Custom roles cannot grant permissions their creator does not hold. Temporary assignments expire automatically.

Permissions are a product feature.

Template matrix: every allowance is further constrained by tenant, campus, relationship and field rules.

ACTION	OWNER	PRINCIPAL	TEACHER	FINANCE	GUARDIAN	SUPPORT
Manage tenant settings	Manage	Delegated	—	—	—	Grant only
View student basics	Scoped	Scoped	Assigned	Minimum	Own child	Grant only
Finalize attendance	Delegated	Approve	Assigned	—	View own	—
Publish final results	Delegated	Approve	Submit	—	View own	—
Post fee receipt	Delegated	View	—	Manage	View own	—
Approve refund	If separate	If delegated	—	If separate	Request	—
Read clinical/case notes	Explicit only	Explicit only	Care alert	—	Reviewed	—
Read salary/bank fields	Explicit HR	Explicit HR	Own	Explicit HR	—	—
Bulk export child data	Step-up	Scoped	Restricted	Finance only	Own child	Grant only

Enforcement contract

Server action permission + row scope + field scope + current relationship. Re-check on asynchronous execution and file download; log sensitive operations.

Navigation with clear boundaries.

Desktop favors operational density. Mobile favors one child, one class or one task at a time.

01

School workspace

Overview • Admissions • Students • Academics • Timetable • Attendance • Learning • Examinations • Fees • Payments

02

People & operations

Staff • Payroll • Messages • Meetings • Transport • Library • Inventory • Health • Visitors • Helpdesk • Reports

03

Administration

School & campuses • Academic years • Roles • Imports • Integrations • Audit log • Branding • Subscription • Privacy

04

Family & student

Today • Children • Timetable • Homework • Results • Attendance • Fees • Messages • Requests • Profile

05

SaaS control plane

Tenants • Plans & usage • Billing • Onboarding • Service health • Support grants • Jobs • Incidents • Feature flags

Every screen preserves current tenant, campus and academic year. Search obeys the same permissions as the underlying records.

Five journeys that must work end to end.

The exception path receives the same implementation attention as the happy path.

01 / A school goes live

Verify owner › **Configure year** › **Import & review** › **Invite users** › **Readiness check**

Recovery: invalid import rows stay in quarantine; activation waits for required setup.

02 / A pupil joins

Enquiry › **Application** › **Offer / seat** › **Verified deposit** › **Enrol & invite**

Recovery: expired offers and full classes route to review or waitlist.

03 / A school day starts

Open roster › **Mark & review** › **Submit register** › **Notify absence** › **Correct / audit**

Recovery: offline edits sync with conflict resolution; unmarked never becomes absent.

04 / A family pays

Select invoice › **Hosted payment** › **Verify capture** › **Post receipt** › **Reconcile bank**

Recovery: delayed webhooks show pending; retries cannot duplicate receipts.

05 / A year closes

Finalize results › **Approve reports** › **Reconcile fees** › **Preview promotion** › **Commit new year**

Recovery: exceptions remain in review; prior reports and enrolments stay intact.

A SaaS business, not just a school portal.

Packaging below is a proposed commercial model. Validate willingness to pay before setting prices.

CAPABILITY	CORE	GROWTH	ENTERPRISE
Student records, attendance, learning, exams	Included	Included	Included
Fees, parent portal, basic reporting	Included	Included	Included
Transport, library, inventory, payroll	Optional add-ons	Included / agreed limits	Contracted
Multi-campus governance / analytics	Single-campus baseline	Multi-campus	Group-wide policies
SSO, dedicated onboarding, advanced SLA	—	Optional	Contracted
Hosting isolation / residency choices	Shared logical isolation	Shared logical isolation	Assessed dedicated option

Billing meter

Define billable active students as a signed-off daily or monthly snapshot of active enrolments, deduplicated across campuses as agreed. Exclude demo, rejected and archived records. Show included allowance, overage and adjustment history.

Lifecycle & cost controls

Trial → active → past due → grace → read-only → closed. Support annual/monthly terms, renewals, credits and cancellation. Meter messages, storage and premium integrations separately. Display the effective date of every plan change.

Keep the school's merchant payments separate from SaaS subscription collections. Never use an unverified frontend flag for entitlements.

Tenant setup & onboarding

Create an isolated school workspace with a verifiable path to its first live day.

TEN / P0

TEN-01

Workspace creation

Verify the owner; capture legal/display names, billing contact, timezone, language and workspace slug. Support school groups, campuses and delegated administrators.

TEN-02

Guided setup

Save progress through academic year, grades, subjects, staff, students, fees and communications. Show dependencies, task owners and completion evidence.

TEN-03

Configuration

Set branding, working week, holidays, number/date formats, document sequences and portal preferences. Preview receipts and report cards.

TEN-04

Invitations & imports

Use expiring single-use invitations. Provide import preview, duplicate review, row errors, checkpoints and reconciliation summaries.

TEN-05

Activation & domains

Check rosters, verified sender, fee test, backup and sample family login. Isolate demo data. Verify custom-domain ownership before DNS/TLS mapping.

TEN-06

Exit lifecycle

Support exports, notice periods, retention holds, cancelled deletion requests and scheduled removal of operational data. Explain backup expiry separately.

PRIMARY OWNER

School owner / SaaS operations

Business rules

Resolve tenant identity server-side. Campus scope never replaces tenant isolation. Sandbox actions cannot message real families.

Acceptance check

Two schools with identical admission numbers see only their own records, files, search and exports. Retried setup creates one tenant.

SCREEN REFERENCES

S01, S02, S19

Identity, roles & permissions

Authorize actions through role, assignment, relationship and field sensitivity.

IAM / P0

IAM-01

Authentication

Offer verified email/phone sign-in, secure recovery and session management. Require MFA for privileged roles; provide recovery codes. Add enterprise SSO in P2.

IAM-02

Membership model

Separate login accounts from staff, student and guardian profiles. Support people in several schools, staff across campuses and guardians with several children.

IAM-03

Authorization

Check tenant, campus, role, academic year, class assignment and guardian relationship on APIs, exports, jobs and downloads. Deny by default.

IAM-04

Role templates

Ship owner, principal, registrar, teacher, finance, HR, librarian, transport, counsellor, guardian, student and support roles. Provide bounded custom roles and effective-access preview.

IAM-05

Delegation & approval

Time-limit substitute access. Require step-up verification and independent approval for refunds, large waivers, payroll release, report reissue and bulk exports.

IAM-06

Lifecycle

Handle expired invites, staff departure, custody changes and shared-device logout. Revoke sessions on access changes. Recovery responses must avoid account enumeration.

PRIMARY OWNER

Owner / security administrator

Business rules

A role alone does not establish child access. Support has no default student-data access; grants require school authorization, scope and expiry.

Acceptance check

A teacher cannot fetch another section by changing an ID. Revoked guardians lose access immediately. A requester cannot approve their own refund.

SCREEN REFERENCES

S01, S18

Admissions & enquiry CRM

Convert enquiries into capacity-checked enrolments with complete application history.

ADM / P0

ADM-01

Intake

Capture web, walk-in, phone and referral enquiries with source, target grade, owner and follow-up. Flag possible duplicates without automatically merging families.

ADM-02

Applications

Build versioned grade-specific forms with conditional fields, saved drafts, document checklists, notices and accessible public links.

ADM-03

Pipeline

Track enquiry, applied, review, assessment, offered, waitlisted, accepted and withdrawn stages, with tasks, history and decision reasons.

ADM-04

Assessment & offers

Book interviews with capacity checks; capture rubrics and restricted notes. Issue versioned offers with expiry, fee terms and outstanding documents.

ADM-05

Seats & conversion

Apply school-approved allocation policy. Atomically reserve seats, verify deposits and convert applicants into student, guardian and enrolment records once.

ADM-06

Reporting

Show source performance, conversion, time in stage, incomplete applications and grade capacity. Export reviewed application packets with scoped access.

PRIMARY OWNER

Admissions team / principal

Business rules

Check offer expiry and capacity at acceptance. Applicant retention differs from permanent academic-record retention. Admission criteria are configurable.

Acceptance check

Concurrent acceptances for the last seat yield one enrolment and one clear capacity error. Repeated conversion cannot create duplicate students.

SCREEN REFERENCES

S03

Student information & lifecycle

Keep reliable records across family changes, years, transfers and graduation.

SIS / P0

SIS-01

Student master

Maintain admission number, preferred/legal name, birth date, photo, address and emergency contacts. Collect additional identifiers only for a defined approved purpose.

SIS-02

Family relationships

Track guardian role, communication preference, financial responsibility, emergency priority, authorized pickup and contact restrictions. Verify relationship changes.

SIS-03

Enrolment history

Store effective-dated campus, year, grade, section, roll number, house and electives. Preserve prior placements when classes change.

SIS-04

Documents

Scan/version uploads; track verification, expiry and download audit. Generate ID cards, bonafide/study and transfer certificates from approved templates.

SIS-05

Promotion & rollover

Preview promotion, retention, transfers and graduation. Copy selected settings, reconcile fee carry-forward and commit an approved batch with recovery planning.

SIS-06

Corrections & alumni

Require reason/review for identity changes and duplicate merges. Preserve source references. Separate opt-in alumni engagement from retained academic records.

PRIMARY OWNER

Registrar / class teacher

Business rules

Never overwrite historical enrolments or issued reports. Custody restrictions apply to messages, files and search. Admission-number corrections are audited.

Acceptance check

Rollover changes current classes while preserving last year’s report. A section transfer changes future rosters without rewriting past attendance.

SCREEN REFERENCES

S04, S05

Academic setup & calendars

Support varied curricula, grade structures and school-year conventions.

ACA / P0

ACA-01

Academic periods

Define years, terms, assessments, working days, holidays and campus exceptions. Lock completed periods with controlled reopening.

ACA-02

Grades & groups

Manage sections, capacities, class teachers, houses, composite teaching groups and cross-section electives.

ACA-03

Subjects

Maintain codes, language options, compulsory/elective rules, practical components and grade-specific offerings.

ACA-04

Board profiles

Version school-validated configurations for CBSE, CISCE and state-board needs. Use configurable rules and templates without claiming board certification.

ACA-05

Teacher allocations

Assign subject/class responsibilities, workloads and effective dates. Retain assignment history for reporting and permissions.

ACA-06

Publishing & cloning

Publish calendars and closure updates. Clone a draft year, compare differences, check missing allocations and require coordinator review.

PRIMARY OWNER

Academic coordinator

Business rules

Do not hardcode April-to-March for every school. Calendar changes flag affected attendance, timetable and billing rules.

Acceptance check

Two schools can use different practical weightings, term structures and language electives without affecting each other.

SCREEN REFERENCES

S07, S08

Timetable & substitutions

Publish workable schedules across teachers, rooms, sections and resources.

TIM / P0 manual / P1 solver

TIM-01

Period templates

Configure periods, breaks, assemblies, rotating weeks and special days, with distinct primary/senior templates.

TIM-02

Scheduling

Offer weekly editing, collision checks, workloads, missing periods and accessible keyboard/list alternatives.

TIM-03

Constraints

Separate hard rules such as availability from soft preferences such as balanced subjects. Explain every conflict precisely.

TIM-04

Versions

Draft, compare, approve and publish with effective dates. Preserve the schedule used for historical attendance.

TIM-05

Substitutions

Find eligible available teachers, record acceptance, notify affected classes and grant only temporary session access.

TIM-06

Generation & sharing

In P1 generate candidate schedules with explanations and locked slots. Require review. Offer teacher/class/room views, print and revocable calendar feeds.

PRIMARY OWNER

Timetable coordinator

Business rules

Hard conflicts block publication. A solver must report an infeasible schedule rather than silently break a hard rule.

Acceptance check

One teacher cannot occupy concurrent classes. Published substitutions update both schedules and temporary access correctly.

SCREEN REFERENCES

S07

Attendance & leave

Make attendance fast to record, safe to correct and accurate to report.

ATT / P0

ATT-01

Registers

Build daily/period rosters from effective enrolments and calendars. Distinguish present, absent, late, excused, unmarked and not scheduled.

ATT-02

Fast marking

Provide touch/keyboard entry, all-present with review, notes, draft saving and an unmarked count before submission.

ATT-03

Offline capture

Queue minimal records on approved devices with expiry and visible pending sync. Use server versions to detect conflicts.

ATT-04

Corrections

Apply deadlines, locks, correction reasons and approvals. Retain before/after values, actor and server receipt time.

ATT-05

Leave & alerts

Accept guardian leave requests with limited attachments. Delay finalized absence alerts; cancel queued alerts when mistakes are corrected.

ATT-06

Analysis

Show register completion, trends and policy-based absence thresholds. Explain denominator exclusions for holidays, future and pre-admission dates.

PRIMARY OWNER

Teachers / attendance officer

Business rules

Unmarked is not absent. Replayed offline events cannot overwrite a newer approved correction. Device time is not the authoritative sequence.

Acceptance check

Conflicting device edits require resolution. Withdrawn pupils remain in history but leave future rosters. Duplicate sync produces one change.

SCREEN REFERENCES

S06, S16

Teaching, homework & learning

Connect lessons, assignments, submissions and deliberately released feedback.

LMS / P0 basics / P1 advanced

LMS-01

Lesson planning

Plan units, objectives, resources and syllabus coverage with coordinator review and reusable school-scoped plans.

LMS-02

Assignments

Set instructions, attachments, rubrics, recipients, due dates and submission types. Draft, schedule, publish, revise and close.

LMS-03

Submission

Provide text/files, progress, scanning, drafts, late flags, resubmissions and individual extensions.

LMS-04

Marking

Support rubric scores, private comments, annotated work and return-for-rework. Release scores explicitly rather than while marking drafts.

LMS-05

Resources & quizzes

Manage access-controlled resources. In P1 add formative quizzes, question banks, objective auto-marking, attempts and accommodations.

LMS-06

Progress & sessions

Show missing work and coverage without public student ranking. Add approved virtual-class links/recordings in P1 with sharing and retention controls.

PRIMARY OWNER

Teachers / students

Business rules

Published changes notify recipients. Retain submission versions. Teachers review automated marking; high-impact decisions are not fully automated.

Acceptance check

An extension changes one student’s deadline. Draft scores remain private until release. Students cannot download one another’s submissions.

SCREEN REFERENCES

S08, S17

Examinations & report cards

Generate approved, reproducible results and traceable report-card versions.

EXM / P0

EXM-01

Assessment rules

Configure components, maxima, pass rules, weightings, grade bands, rounding and absent/exempt codes. Version by year/grade.

EXM-02

Logistics

Schedule rooms/invigilators, accommodations, seating and admit cards. Detect student, room and teacher collisions.

EXM-03

Marks entry

Restrict entry to assigned subjects. Validate imports, ranges and required fields; distinguish zero, absent, exempt and missing.

EXM-04

Moderation

Use draft, submitted, reviewed, approved and published states. Lock submissions; require reason and authorization to reopen.

EXM-05

Report generation

Create branded multilingual-ready PDFs with grades, competencies, attendance, remarks and signatories. Snapshot values and calculation rules.

EXM-06

Publication & appeals

Release by cohort on schedule. Provide secure family access, review requests and controlled reissue retaining prior versions and reasons.

PRIMARY OWNER

Exam coordinator / teachers

Business rules

Required missing marks block publication. Scheme changes do not alter issued reports. Fee arrears must not automatically hide results.

Acceptance check

Screen and PDF use the same rounding. Reissue creates a new version and publication record while the original remains reproducible.

SCREEN REFERENCES

S09

Fees, concessions & receivables

Create correct charges, transparent balances and controlled adjustments.

FEE / P0

FEE-01

Plans

Configure tuition, admission, transport, hostel and optional heads, instalments, due dates, effective dates and explicit late-fee rules.

FEE-02

Allocations

Assign by enrolment/service. Preview proration, joins, withdrawals, sibling benefits and opening balances before posting.

FEE-03

Concessions

Manage scholarships, waivers and discounts with eligibility, validity, caps, funding and approval. Prevent incompatible stacking.

FEE-04

Invoices

Number line-item invoices; support partial payments, credit notes, cancellation and student/family ledgers. Store money in minor units.

FEE-05

Collections

Record cash, cheque, transfer and verified online payments. Track cheque pending/cleared/bounced; reconcile cashier sessions and receipts.

FEE-06

Dues & refunds

Show aging, installment plans and disputes; pause contested reminders. Apply overpayments as credits. Refund only eligible unrefunded payments.

PRIMARY OWNER

Finance officer / approver

Business rules

Reverse posted entries instead of editing them. Student fees and SaaS invoices use separate accounts/sequences. Tax treatment requires accounting review.

Acceptance check

Partial payment reduces the correct balance. Credit notes post once. Cashiers cannot approve their own waivers or refunds.

SCREEN REFERENCES

S10, S15

Payments & reconciliation

Match student balances, provider records and bank settlements without double posting.

PAY / P0

PAY-01

Merchant setup

Use school-owned approved gateway accounts; segregate tenant and test/live credentials. Track merchant onboarding and secret rotation.

PAY-02

Checkout

Use provider-hosted UPI/cards/netbanking where available. Bind orders to amount, currency, invoice and payer; expire stale attempts.

PAY-03

Verification

Treat redirects as pending evidence. Verify raw-byte webhook signatures, merchant context and provider status before issuing confirmed receipts.

PAY-04

Durable processing

Persist verified events, deduplicate event IDs and enforce unique business postings. Handle retries and out-of-order events with explicit states.

PAY-05

Reconciliation

Match captures, receipt allocations, refunds, gateway charges and bank settlements. Assign unmatched exceptions and run daily reconciliation.

PAY-06

Recovery & disputes

Poll unresolved attempts, safely retry jobs and audit replay. Track chargebacks/refund deadlines and show pending states without repeated-pay prompts.

PRIMARY OWNER

Finance / payment operations

Business rules

Captured, receipted and settled are distinct. Settlement charges cannot reduce the student's paid amount. Confirm provider contracts; see [4].

Acceptance check

Ten concurrent duplicate capture events create one receipt/posting. Late authorization cannot move a captured payment backwards.

SCREEN REFERENCES

S11, S15

Accounting & financial controls

Provide auditable operational books and a controlled accounting handoff.

FIN / P1

FIN-01

Subledger

Use balanced double-entry postings for receivables, collections, concessions, refunds and gateway clearing. Agree mappings with accountants.

FIN-02

Cash controls

Track opening float, shifts, mode totals, variance explanations, supervisor sign-off and protected receipt sequences.

FIN-03

Expenses

Manage vendors, bills, categories, cost centres, attachments and approvals. Capture settlement references without putting credentials in notes.

FIN-04

Budgets

Track department/campus budgets, commitments and actual spend. Route exceptional purchases and threshold breaches.

FIN-05

Bank reconciliation

Import statements with duplicate detection, suggested matches, source provenance, unresolved items and reconciliation locks.

FIN-06

Close & export

Restrict backdating, approve adjustments and close periods. Reconcile counts/totals when exporting to accounting packages. Validate statutory filing separately.

PRIMARY OWNER

Accountant / school management

Business rules

Every journal balances within one currency. Reopening a period requires a named approver.

Acceptance check

Refunds and gateway charges produce balanced entries. Ordinary users cannot post in closed periods. Export retries do not duplicate postings.

SCREEN REFERENCES

S10, S11

Staff, leave & payroll

Manage employment, availability and a reviewable payroll cycle.

HRM / P0 staff / P1 payroll

HRM-01

Staff records

Maintain employee IDs, roles, qualifications, contracts, join/exit dates and emergency contacts. Restrict bank, identity and salary fields.

HRM-02

Leave & attendance

Set shifts, balances, requests and approvals. Reconcile approved leave with payroll and timetable substitutions.

HRM-03

Pay structures

Version allowances, deductions, reimbursements, arrears and effective dates. Separate recurring and one-time components.

HRM-04

Payroll runs

Draft with exceptions and comparisons, independently approve, lock, disburse and reconcile failed payments.

HRM-05

Payslips & local rules

Publish secure versioned payslips. Configure applicable PF, ESI, TDS and professional tax using current reviewed inputs or a payroll provider.

HRM-06

Lifecycle

Manage onboarding, document expiry, training, appraisal, clearance and final settlement. Reassign duties and revoke access at departure.

PRIMARY OWNER

HR / payroll approver

Business rules

No statutory rates are prescribed here. General administrators do not automatically see salaries. Published payroll is corrected through a new version.

Acceptance check

Retroactive salary changes create visible arrears. Teachers retrieve only their own payslips. Preparers cannot release their own run.

SCREEN REFERENCES

S12

Communication & engagement

Deliver targeted messages with clear consent, recipients and delivery status.

COM / P0

COM-01

Notices

Compose categorized notices, translations, attachments and expiry. Preview recipients, schedule sends and approve sensitive broadcasts.

COM-02

Channels

Launch in-app/email, then approved SMS/WhatsApp. Distinguish provider acceptance, delivery, failure and read status.

COM-03

Preferences

Respect purpose, consent, verified contacts and guardian restrictions. Apply quiet hours with an authorized emergency override.

COM-04

Templates

Version templates/approvals. Support provider-required Indian SMS sender/template registration and quotas.

COM-05

Conversations & meetings

Offer moderated family-school conversations, office hours, escalation and safe attachments. Book parent-teacher slots with capacity/rescheduling.

COM-06

Feedback & costs

Collect surveys, acknowledgement and consent. Estimate billable sends, deduplicate recipients, save audience snapshots and handle applicable opt-outs.

PRIMARY OWNER

School office / teachers

Business rules

Read receipts do not prove consent. Contact changes recheck verification and permission. Emergency override is recorded.

Acceptance check

A section notice reaches only authorized families. Correcting absence cancels its unsent alert. Failed delivery stays visible for recovery.

SCREEN REFERENCES

S13

Parent & student portals

Give families a clear daily view and safe paths to complete school tasks.

FAM / P0

FAM-01

Home

Show today’s timetable, attendance, homework, notices and next action. Keep selected school/child visible across child switching.

FAM-02

Fees

Explain charges, concessions, allocations, disputes and pending payments. Provide hosted checkout and secure receipts.

FAM-03

Learning

Students access resources, submissions and released feedback. Guardians receive only school-approved learning information.

FAM-04

Requests

Handle leave, contact updates, certificates, pickup and meetings with status, expected response and history.

FAM-05

Accessible mobile

Provide language preference, readable type, keyboard/screen-reader support, large targets, low-bandwidth views and file-size hints.

FAM-06

Safety

Use minimal push content, authenticated deep links, verified guardian linking, age-appropriate permissions and expiring caches.

PRIMARY OWNER

Guardians / students

Business rules

Student and guardian accounts are distinct. Switching children rechecks access. Unpaid fees do not automatically disable education access.

Acceptance check

A guardian pays one child’s invoice and retrieves its receipt without other-family data. Revoked relationships lose inbox and file access.

SCREEN REFERENCES

S15, S17

Transport & student pickup

Coordinate routes, riders and handover with honest tracking reliability.

TRN / P1

TRN-01

Fleet

Track vehicles, capacity, drivers, attendants and permit/insurance/fitness expiry. Flag or block invalid assignments under school policy.

TRN-02

Routes

Configure stops, directions, schedules, academic-day applicability and effective-dated riders. Check capacity and pickup/drop windows.

TRN-03

Trips

Create daily manifests, record boarding/alighting and reconcile missing riders. Provide minimal offline manifests to attendants.

TRN-04

Location

Integrate approved GPS; show last update, signal loss and approximate ETA. Restrict tracking to active trips and authorized families.

TRN-05

Changes & incidents

Approve one-day stop/pickup changes, verify handover and notify attendants. Route breakdowns, delays and emergencies through a contact tree.

TRN-06

Fees & reports

Link subscriptions to fees. Report utilization, punctuality, missing boardings and expiry risks with restricted location history.

PRIMARY OWNER

Transport manager / attendant

Business rules

GPS is advisory, not guaranteed safety monitoring. Retention requires a justified duration. Never expose unrelated children’s locations.

Acceptance check

Stale GPS visibly loses its ETA. Concurrent assignments cannot overfill a vehicle. Unauthorized families cannot access the trip.

SCREEN REFERENCES

S14

Library & circulation

Track titles and individual copies from acquisition to withdrawal.

LIB / P1

LIB-01

Catalogue

Separate bibliographic titles from accessioned copies; store author, language, ISBN, shelf, category and acquisition.

LIB-02

Loans

Issue, renew and return via barcode/QR; check borrower eligibility, limits, due dates and reservations.

LIB-03

Holds

Manage queues, pickup expiry and notifications. Prevent simultaneous loans of one copy.

LIB-04

Fines

Configure grace, caps, damage/loss and approved waivers. Integrate approved charges with finance without duplicates.

LIB-05

Stock & digital

Scan stocktakes, reconcile missing items and preserve history of withdrawn copies. Link licensed resources with access and expiry.

LIB-06

Reporting

Show overdue loans, circulation, stock value and borrower clearance at transfer/graduation with controlled borrowing-history access.

PRIMARY OWNER

Librarian

Business rules

At most one open loan per copy. Fines require policy authorization. Digital material needs appropriate school permission.

Acceptance check

Concurrent issues create one open loan. Returning a lost book triggers fine/refund review rather than an automatic refund.

SCREEN REFERENCES

S20

Inventory, purchasing & assets

Control purchasing and stock through auditable movements and approvals.

INV / P1

INV-01

Items

Manage consumables, serialized assets, units, locations, reorder levels, campus ownership and relevant lots/expiry.

INV-02

Purchasing

Handle requisition, budget check, quotations, approval, purchase order and cancellation using policy thresholds.

INV-03

Receiving

Record partial deliveries, rejection, shortage, vendor return and invoice matching with evidence.

INV-04

Movements

Issue, return and transfer with recipient/purpose. Block negative stock except under explicitly approved policy.

INV-05

Assets

Assign devices, furniture and lab equipment to people/rooms. Track condition, warranty, repair, custody and disposal.

INV-06

Audits

Count stock, investigate variances and approve adjustments. Report vendors, commitments and consumption; export reviewed accounting entries.

PRIMARY OWNER

Storekeeper / procurement

Business rules

Separate requester and approver for controlled spend. Derive balances from movements rather than editable totals.

Acceptance check

A partial receipt leaves an order partly open. Duplicate receipt imports cannot inflate stock. Unauthorized disposal is rejected.

SCREEN REFERENCES

S21

Health, wellbeing & safeguarding

Support student care while isolating especially sensitive records.

HLT / P0 essentials / P1 casework

HLT-01

Medical profiles

Collect necessary allergies, care instructions and emergency contacts. Separate detailed medical records from the general profile.

HLT-02

Clinic visits

Record observations, action, referral and notification. Medication workflows require documented authorization and trained staff.

HLT-03

Care permissions

Show assigned teachers only essential alerts. Treatment/counselling notes require explicit role and purpose.

HLT-04

Incidents

Capture concerns, witnesses and restricted evidence. Assign severity, owner, follow-up and escalation under school-approved procedures.

HLT-05

Case privacy

Use sealed permissions, redaction and restricted exports. Do not automatically share case notes through family portals.

HLT-06

Support & emergencies

Track accommodations/reviews with approved staff. Provide minimal emergency rosters and manual fallback; audit sensitive reads/exports.

PRIMARY OWNER

Nurse / safeguarding lead

Business rules

Software does not diagnose or replace safeguarding judgement. Schools approve disclosure, escalation and retention procedures.

Acceptance check

Finance cannot discover counselling notes via search, API or export. Teachers see necessary allergy instructions without detailed medical files.

SCREEN REFERENCES

S05 restricted panel

Front office, events & helpdesk

Make campus visitors, requests and events accountable and easy to follow.

OPS / P1

OPS-01

Visitors

Pre-register, verify hosts, issue expiring passes and log entry/exit. Minimize identity retention.

OPS-02

Pickup

Manage authorized people, one-time changes, gate verification and exceptions. Record staff releasing the child and the recipient.

OPS-03

Events

Set capacity, fees, schedule, consent, participants and emergency contacts. Manage waitlists, cancellation and refunds.

OPS-04

Facilities

Book rooms, grounds and equipment with collision checks, access rules and maintenance blackouts.

OPS-05

Helpdesk & office

Route IT, maintenance, academic and billing tickets with owner, priority and service target. Separate internal notes. Process certificates and dispatch.

OPS-06

Public pages

In P2 offer school pages, enquiry forms and announcements with consented media, moderation, accessible templates and spam protection.

PRIMARY OWNER

Office / operations manager

Business rules

Visitor approval is not pickup authorization. Public pages never expose private records. Ticket notifications respect participant scope.

Acceptance check

Expired passes require renewal. Concurrent bookings cannot occupy one room. Internal notes remain hidden from families.

SCREEN REFERENCES

S13, S22

Hostel, meals & enrichment

Add residential and enrichment operations only where required.

EXT / P2 optional

EXT-01

Hostel

Model residences, rooms, beds, approved allocation policies, effective dates, waitlists and moves.

EXT-02

Residential attendance

Record roll call, overnight leave, guardian approval, gate movement and safe return; escalate unexplained absence.

EXT-03

Services

Track inspections, belongings, room issues, deposits and boarding fees. Limit wardens to relevant residents.

EXT-04

Meals

Plan menus, dietary instructions, headcounts and stock. Use controlled meal credits with explicit accounting/refunds; review wallet-like features separately.

EXT-05

Activities

Manage clubs, sports, teams, houses, achievements, instructors, consent, capacity and optional charges.

EXT-06

Community

Offer opt-in alumni events and authorized donations with separate receipts/accounts and engagement consent.

PRIMARY OWNER

Wardens / activity leads

Business rules

Residential leave does not change classroom attendance automatically. Staff verify dietary instructions. Optional modules inherit tenant/access controls.

Acceptance check

One bed cannot host overlapping stays. Cancelled paid activities follow refund approval. Wardens cannot access other residences.

SCREEN REFERENCES

Reuse S04, S06, S21

Analytics & reporting

Make metrics traceable, permission-aware and useful for decisions.

ANA / P0 essentials / P1 advanced

ANA-01

Dashboards

Show attendance, admissions, fees, learning, staffing and services with role-appropriate campus/year/date filters.

ANA-02

Definitions

Show formula, population, exclusions, source and refresh time. Distinguish invoiced fees, collections, capture and settlement.

ANA-03

Drilldown

Preserve filters and access rules. Apply privacy restrictions for sensitive small cohorts.

ANA-04

Report library

Provide registers, aging, concessions, cash close, results, student strength, staff leave and audit reports with versioned layouts.

ANA-05

Exports

Generate large files asynchronously with expiry/counts/audit. Recheck scheduled recipients and authorization at download.

ANA-06

Assistive insight

In P2 add reviewed summaries and explainable exception suggestions. Do not automate admissions, discipline, final grading or eligibility decisions.

PRIMARY OWNER

Principal / group leadership

Business rules

Group visibility is explicitly granted. Avoid unnecessary direct student identifiers in analytics. Totals reconcile to filtered source records.

Acceptance check

Campus users cannot expand scope with crafted filters. A revoked user cannot download a previously generated report.

SCREEN REFERENCES

S02, S23

Migration, integrations & automation

Move records and connect providers without losing traceability.

INT / P0 imports / P1 ecosystem

INT-01

Templates

Support CSV/XLSX mapping, encoding/date checks, duplicates, dry runs and row errors. Reconcile student counts and opening balances.

INT-02

Import jobs

Checkpoint chunks, quarantine invalid rows, preserve external IDs and produce manifests. Allow controlled rollback before activation.

INT-03

APIs

Use versioned scoped APIs with pagination, limits, idempotency and deprecation. Derive tenant scope from validated membership.

INT-04

Events

Publish signed minimal webhooks with IDs, retries, logs and audited replay. Restrict event types and destinations.

INT-05

Connectors

Prioritize payments, messaging, accounting and calendars. Separately evaluate GPS, SSO, staff biometric devices and learning platforms.

INT-06

Automation

Use allowed event-condition-action rules with dry run, caps, loop protection and approval for sensitive changes. Track provider health/credentials.

PRIMARY OWNER

School IT / integrations

Business rules

Government or board API access is not assumed. UDISE+ and board reporting start as reviewed export mappings pending a supported authorized interface.

Acceptance check

Unchanged external IDs update/skip under the selected mode, with reconciled totals. Provider outages produce visible retry/manual states.

SCREEN REFERENCES

S18, S22

SaaS control plane & support

Operate commercial subscriptions, tenant health, support and exits.

PLT / P0 core / P1 enterprise

PLT-01

Console

Show tenant health, lifecycle, plan, campuses, usage and onboarding without child data by default.

PLT-02

Subscriptions

Track trial, active, past-due, grace, read-only, cancelled and closed states with notices/effective dates. Separate school billing from student fees.

PLT-03

Entitlements

Enforce features/quotas server-side. Record override owner, reason and expiry. Preserve historical reading/export after downgrade under contract.

PLT-04

Usage & invoices

Snapshot billable active students under a published definition; reconcile adjustments, invoices, renewals and credits.

PLT-05

Support grants

Require a ticket and school-approved scoped session with expiry/banner. Log actions and prohibit invisible permanent impersonation.

PLT-06

Operations & exit

Monitor queues, providers, hotspots and storage. Use staged flags/kill switches. Export, honor holds and delete operational copies under policy.

PRIMARY OWNER

SaaS operations / customer success

Business rules

Subscription failure does not instantly remove critical-record access. Backup expiry is separate from live deletion.

Acceptance check

Retrying a usage snapshot yields one invoice. Expired support grants cannot query students or download earlier exports.

SCREEN REFERENCES

S19, S22

Architecture that can grow with the product.

Recommended starting point: a modular application with explicit domain boundaries and managed infrastructure.

Responsive clients

Admin web • teacher web/PWA • family portal • operator console

↓

Edge & identity

CDN / TLS / rate limits • verified sessions • tenant/campus context

↓

Application modules

SIS & admissions • academics • finance • operations • SaaS control plane

↓

Durable async work

Transactional outbox • job workers • provider inbox • retries / dead-letter queue

↓

Data & integrations

PostgreSQL • private object storage • cache • payment / messaging / GPS adapters

Suggested implementation

TypeScript across frontend and backend; React-based responsive UI; PostgreSQL transactional store; private object storage; durable queue; OpenAPI contracts. Select maintained framework versions during build.

Why this structure

One deployable application initially keeps cross-domain transactions manageable. Isolate modules in code and data access. Extract services only when scale, team ownership or failure isolation justifies it.

External dependencies

Use adapters for payment, communications, storage and GPS. Vendor failures must not corrupt school records. Infrastructure/provider choice remains an implementation decision.

This is a proposed architecture, not an installed stack. No particular cloud, database vendor or payment provider is required.

Tenant isolation is enforced at every layer.

Recommended default: shared PostgreSQL with tenant-aware keys, policy enforcement and a non-privileged runtime role.

LAYER	REQUIRED CONTROL	VERIFICATION
Request context	Derive tenant from authenticated membership; validate campus scope.	Reject crafted tenant/campus IDs.
Database	tenant_id on scoped rows; composite foreign keys; RLS as defense in depth.	Attempt cross-tenant read/write/joins.
Connection pooling	Set transaction-local scope; reset context; no owner/BYPASSRLS runtime.	Reuse pooled connections across tenants.
Object storage	Private objects, tenant paths, short-lived links after authorization.	Try guessed paths and expired links.
Cache / search	Include tenant and permission-sensitive scope in keys/index queries.	Check cache/search results after role changes.
Jobs / exports	Carry explicit tenant + actor; revalidate scope when executing/downloading.	Revoke permissions while job is queued.
Control plane	Separate operator roles and data access; time-limited school grants.	Expired support grant fails everywhere.
Analytics / backups	Scoped aggregation; restricted restore tooling; audited data movement.	Reconcile tenant export and recovery scope.

Database caveat [5]

Row-level policies supplement application authorization. Database owners, privileged jobs and maintenance paths can bypass ordinary policy behavior; inventory and test these paths explicitly.

A durable model of a changing school.

Accounts, people and enrolments are separate. Academic and financial history is preserved.

DOMAIN	CORE ENTITIES	RELATIONSHIP / SOURCE OF TRUTH
Tenancy	Tenant, Campus, AcademicYear, Membership, RoleGrant	Tenant owns campuses; grants carry scope/effective dates.
People	Account, Student, Guardian, StudentGuardian, Employee	Guardian ↔ student is verified many-to-many; account is optional.
School structure	Grade, Section, Subject, TeachingAssignment, Enrolment	Student → many historical enrolments; section belongs to year/campus.
Admissions	Enquiry, Application, Offer, SeatReservation	Accepted application → one enrolment conversion.
Teaching	TimetableVersion, Session, Lesson, Assignment, Submission	Session uses effective roster; submissions have versions.
Assessment	Exam, Component, MarkEntry, SchemeVersion, ReportIssue	Issued report snapshots scheme and approved values.
Attendance	Register, AttendanceEntry, LeaveRequest, Correction	Unique student + session; corrections retain audit history.
Money	Invoice, Line, CreditNote, Payment, Allocation, Refund, Journal	Payment ↔ invoice via allocation; reversals are explicit.
Operations	Route, Trip, Loan, Asset, StockMovement, Ticket, CareCase	Each has scoped owner, status, history and service permissions.
Platform	Subscription, UsageSnapshot, WebhookInbox, Outbox, AuditEvent	School subscription books are distinct from student-fee books.

Use opaque stable IDs. Attach tenant_id to scoped rows and include it in cross-table foreign keys to prevent cross-tenant references.

Data integrity, lifecycle & retention.

Minimum constraints belong in the database as well as application validation.

CONTRACT	IMPLEMENTATION REQUIREMENT
Identity	Unique tenant + admission number; explicit correction flow. IDs remain stable through profile changes.
Enrolment	Prevent overlapping active placements where school policy forbids them; retain effective dates/history.
Attendance	Unique tenant + student + session; version number prevents lost updates; no default absence for missing rows.
Money	Integer minor units + ISO currency; allocation totals cannot exceed available balances; refunds cannot exceed eligibility.
Payment processing	Unique tenant + provider + merchant + event ID; unique verified payment posting; atomic receipt + ledger + outbox.
Capacity	Atomic seat/bed reservation and single open library loan per copy; prevent simultaneous resource overbooking.
Publication	Immutable issued report snapshot, template version, signer and checksum; reissue references the previous version.
Audit	Append-only actor, tenant, action, target, server timestamp, reason and safe before/after summary; redact secrets.

Retention decision register

For each record class define purpose, owner, legal/contract basis, retention trigger, duration, hold rules and deletion method. Cover applicant files, school records, finance, health, GPS, messages, audit logs and backups.

Deletion semantics

Use archive for operational lifecycle; use actual deletion/anonymization for approved erasure. Do not treat soft delete as completed erasure. Export and legal holds require explicit review and audit.

API contracts & reliable background jobs.

Illustrative versioned routes. Final schemas belong in OpenAPI with request, response, auth scope and error examples.

ROUTE / ACTION	SCOPE	SAFETY CONTRACT
POST /v1/applications/{id}/accept	admissions.accept	Seat reservation + enrolment transaction; idempotency key.
PUT /v1/registers/{id}/entries	attendance.write	Roster validation; expected version; conflict returns 409.
POST /v1/reports/{id}/publish	results.publish	Required marks resolved; independent approval; immutable issue.
POST /v1/invoices/{id}/payment-attempts	fees.pay / own child	Server amount/currency; provider order; safe retry semantics.
POST /v1/webhooks/payments/{connection}	Provider signature	Raw-body verify; durable inbox; dedupe; async processing.
POST /v1/payments/{id}/refund-requests	refund.request	Eligibility check; separate approval; provider reconciliation.
POST /v1/exports	export.create	Authorized filter snapshot; async job; expiring download.
POST /v1/support-grants	tenant.owner	Ticket, scope, reason, expiry and audit; revocable at any time.

Shared contract

Cursor pagination; bounded filters; stable error codes with field details and correlation ID; consistent time/money formats. Use 401/403, 409, 422 and 429 deliberately. Never expose another tenant’s record existence.

Job contract

Durable queue, attempt count, bounded exponential retry, dead-letter queue and authorized replay. Idempotency keys bind actor/tenant/action/body hash. Use an outbox for post-transaction notifications.

Payments: explicit states, recoverable failures.

A bank movement and an application message are different events. Reconciliation closes the loop.

Capture path

Attempt created › **Checkout pending** › **Provider captured** › **Verified posting** › **Receipt issued**

Settlement path

Captured ledger › **Provider settlement** › **Bank statement** › **Match / exception** › **Close batch**

Failure handling

Browser closes: recover using provider events/status.
Duplicate webhook: acknowledge stored event; no second posting.
Out-of-order event: apply valid transitions only.
Worker crashes: retry the durable inbox transaction.
Amount mismatch: quarantine for review.

Refund & dispute control

Requested → independently approved → provider initiated → pending → completed / failed. Keep the original payment and record a separate reversal only when justified. Track partial refunds and chargebacks independently. Reconcile gross amount, charges, net settlement and bank references.

PAY acceptance: verified receipt + allocation + ledger + outbox commit atomically; family-facing status never relies only on a redirect. [4]

Security, privacy & child data.

Implement these controls from P0. Validate applicable legal obligations and commencement dates before launch.

Application security [2]

Use ASVS 5.0 as a verification catalogue. Require MFA for privileged users, short-lived sessions, secure cookies, CSRF protection, input validation, dependency scanning, secret management and rate limits. Test file uploads for type/size, malware and unauthorized retrieval.

India privacy readiness [3]

Maintain a processing inventory and purpose-specific notices. Implement guardian verification and consent records where applicable, withdrawal handling, correction/erasure requests, grievance routing and breach response. Validate school/vendor responsibilities, exemptions and staged commencement against official material.

Child-safe defaults

No targeted advertising or sale of child data. Minimize identity collection; do not require Aadhaar by default. Use restricted health/custody access, private media, verified guardian links and minimal lock-screen notifications. Prohibit training third-party models on school data without an approved basis and contract.

Governance & evidence

Agree school/vendor data roles, subprocessors, hosting region, access requests, retention and offboarding in contracts. Record policy versions and approval. Encrypt transport/storage; isolate production; redact logs; audit sensitive reads, support access and exports. Rehearse incident triage and notification decisions.

Legal review is a launch dependency. This document specifies product controls; it does not assert that every law is fully in force or that the product is compliant.

Measurable quality requirements.

Proposed engineering acceptance targets; benchmark against a representative pilot dataset before making SLA commitments.

ID	REQUIREMENT	TARGET / TEST CONDITION
NFR-01	Availability	99.9% monthly core-service target; define exclusions and measurement in the SLA.
NFR-02	Interactive performance	p95 API reads < 500 ms, writes < 800 ms excluding provider latency under agreed load.
NFR-03	Web experience	p75 LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1 on representative midrange mobile/4G.
NFR-04	Initial load envelope	Benchmark 100 tenants × 2,000 pupils; 2,000 concurrent sessions; 300 requests/s peak.
NFR-05	Batch processing	10,000-row import/export completes within 10 min in benchmark; visible progress/cancellation.
NFR-06	Recovery	Proposed RPO ≤ 15 min and RTO ≤ 4 h for core data; prove through restore exercises.
NFR-07	Accessibility [1]	WCAG 2.2 AA target; keyboard, screen reader, focus, errors and zoom checks on key journeys.
NFR-08	Consistency	No duplicated receipt, capacity overbooking or lost approved correction under concurrency tests.
NFR-09	Compatibility	Current and prior major Chrome/Edge/Safari/Firefox; responsive widths 360–1920 px.
NFR-10	Localization	Externalized strings; English + Hindi pilot where validated; Unicode names, INR and local dates.

The load envelope is a planning assumption, not a measured capacity claim. Include peak attendance, payment and result-release bursts.

Deploy, observe, restore.

Operational readiness is part of the product scope and has named ownership.

Environments & delivery

Separate local, staging and production accounts/data. Use synthetic test records, reviewed migrations, infrastructure-as-code, protected secrets and CI gates. Deploy with feature flags and gradual tenant rollout. Use backward-compatible expand/migrate/contract schema changes.

Observability

Capture metrics, redacted structured logs and traces with tenant-aware correlation. Monitor latency, error rate, queue age, provider failures, overdue reconciliation, storage and backup age. Define alert ownership, severity and runbooks; avoid student details in alerts.

Backup & recovery

Encrypt scheduled backups and point-in-time recovery. Restrict backup operators and region copies. Rehearse full restoration and tenant-level export recovery. Verify database, files and encryption-key dependencies together. Confirm counts, balances and checksums after restore.

Incident & continuity

Maintain on-call escalation, status communication and evidence preservation. Provide manual attendance/cash-collection fallback with later reconciliation. Disable failing integrations safely. Practice provider outage, accidental deletion, credential compromise and regional failure.

Release rollback is not database rollback. Document forward-fix and data-recovery strategies for each migration; test them before a risky rollout.

A release gate with meaningful evidence.

The following are scenario requirements for implementation, not claims that the application has already passed.

TEST ID	SCENARIO	REQUIRED RESULT
QA-01	Cross-tenant API, search, file and cache access	Denied without leakage; security event recorded.
QA-02	Guardian revoked while session/export is active	API and file access stop; communication audience updated.
QA-03	Concurrent seat, bed, room and copy allocation	Capacity constraint holds; loser gets actionable conflict.
QA-04	Offline attendance vs approved correction	Conflict resolved; no silent overwrite or duplicate alert.
QA-05	Duplicate/out-of-order/tampered payment events	One valid posting; invalid signature rejected; audit preserved.
QA-06	Partial refund, chargeback and settlement mismatch	Balances reconcile; exception owner and status visible.
QA-07	Missing marks, scheme change and report reissue	Publication blocks correctly; earlier report stays intact.
QA-08	Rollover with repeats, transfers and outstanding dues	Correct enrolments, fee carry-forward and historical reports.
QA-09	Import retry, malformed file and duplicate IDs	Safe validation, recoverable progress and reconciled totals.
QA-10	Keyboard, screen reader, zoom and low bandwidth	Key tasks complete with understandable errors and focus.
QA-11	Backup restore + job replay + provider outage	Recovery targets met; no duplicate financial side effects.

Go-live gate: zero unresolved critical/high access or financial defects; signed pilot UAT; restore evidence; reconciled migration; trained owners.

From discovery to a sustainable release.

Illustrative sequencing for a dedicated cross-functional team; estimates require discovery and capacity confirmation.

PHASE	INDICATIVE WINDOW	DELIVERABLE / EXIT GATE
Discovery & design	Weeks 1–3	Pilot interviews, school policy mapping, source data audit, validated prototype and signed P0 backlog.
Platform foundation	Weeks 4–7	Tenancy, IAM, roles, audit, environments, imports, school/year setup and security tests.
School core	Weeks 8–13	Admissions, SIS, manual timetable, attendance, staff directory, teaching and family portal.
Finance & results	Weeks 14–19	Fees, verified payments, reconciliation, examinations/report cards and communications.
Pilot & launch	Weeks 20–23	Migration rehearsals, UAT, accessibility/load checks, restore exercise, training and monitored go-live.
P1 operations	Next 12–16 weeks	Transport, library, purchasing, deeper HR/payroll, accounting, analytics and scheduling solver.
P2 options	Demand-led increments	Hostel, meals, enterprise SSO, public pages, assistive AI and further localization.

Suggested team & costing

Product/domain lead; product designer; technical lead; 3–4 engineers across frontend/backend; QA automation; shared DevOps/security and school/accounting reviewers. Estimate cost as role-months × agreed rates plus hosting, messaging, gateways and support.

Dependencies, decisions & rollout readiness.

Resolve material product assumptions before committing to scope, cost or a customer SLA.

DECISION / RISK	OWNER	REQUIRED EVIDENCE
Pilot school sizes, boards and operating model	Product + school owners	Validated workflows, volumes and signed priority exceptions.
School/vendor data roles, consent and retention	Legal/privacy + school	Reviewed contract, notices, retention register and legal timeline.
Fee, concession, tax, refund and payroll policies	Finance + reviewers	Signed calculation examples and accounting mappings.
Provider access and ongoing unit costs	Engineering + procurement	Sandbox tests, commercial terms, quotas and outage procedures.
Data migration quality / opening balances	Registrar + finance	Reconciled dry run, issue register and school approval.
Budget, staffing and schedule	Delivery lead	Capacity plan and estimates against the requirements register.
Regional rollout and support languages	Customer success	Training materials, language QA and staffed escalation channels.

Critical dependency order

TEN + IAM → SIS + ACA → TIM / ATT / LMS / EXM.
SIS + FEE → PAY → FIN.
IAM + staff records → payroll.
All modules → ANA + platform operations.
Security, audit and accessibility run through every phase.

Definition of done

Reviewed design + business rules; server authorization; meaningful tests; failure/empty/loading states; accessibility; analytics; documentation; support runbook; school acceptance. Feature completion includes recovery and migration behavior.

A visual system with room to breathe.

Design direction: calm operational clarity, warm surfaces, precise typography and restrained violet accents.

Ink
#18243D

Primary
#6553D9

Lavender
#EFEDFF

Canvas
#F6F7FB

Success
#087C67

Warning
#976009

Type & spacing

Segoe UI in this PDF; use an appropriately licensed UI font in production. Screen hierarchy: 28/20/16/14/12 px equivalent; body line-height around 1.45. Use a 4 px grid, 16–24 px card padding, 10–16 px radii and subtle borders.

Reusable components

Workspace switcher, page header, metric card, data table, search/filter bar, status label, tabs, calendar, form field, drawer, modal, stepper, empty state, toast, upload queue, approval timeline and audit history.

Responsive rules

Desktop: persistent navigation, 12-column content grid and dense but readable tables. Tablet: collapsed rail and prioritized columns. Mobile: one-column tasks, bottom navigation and explicit child/class context. Offer list alternatives to wide tables.

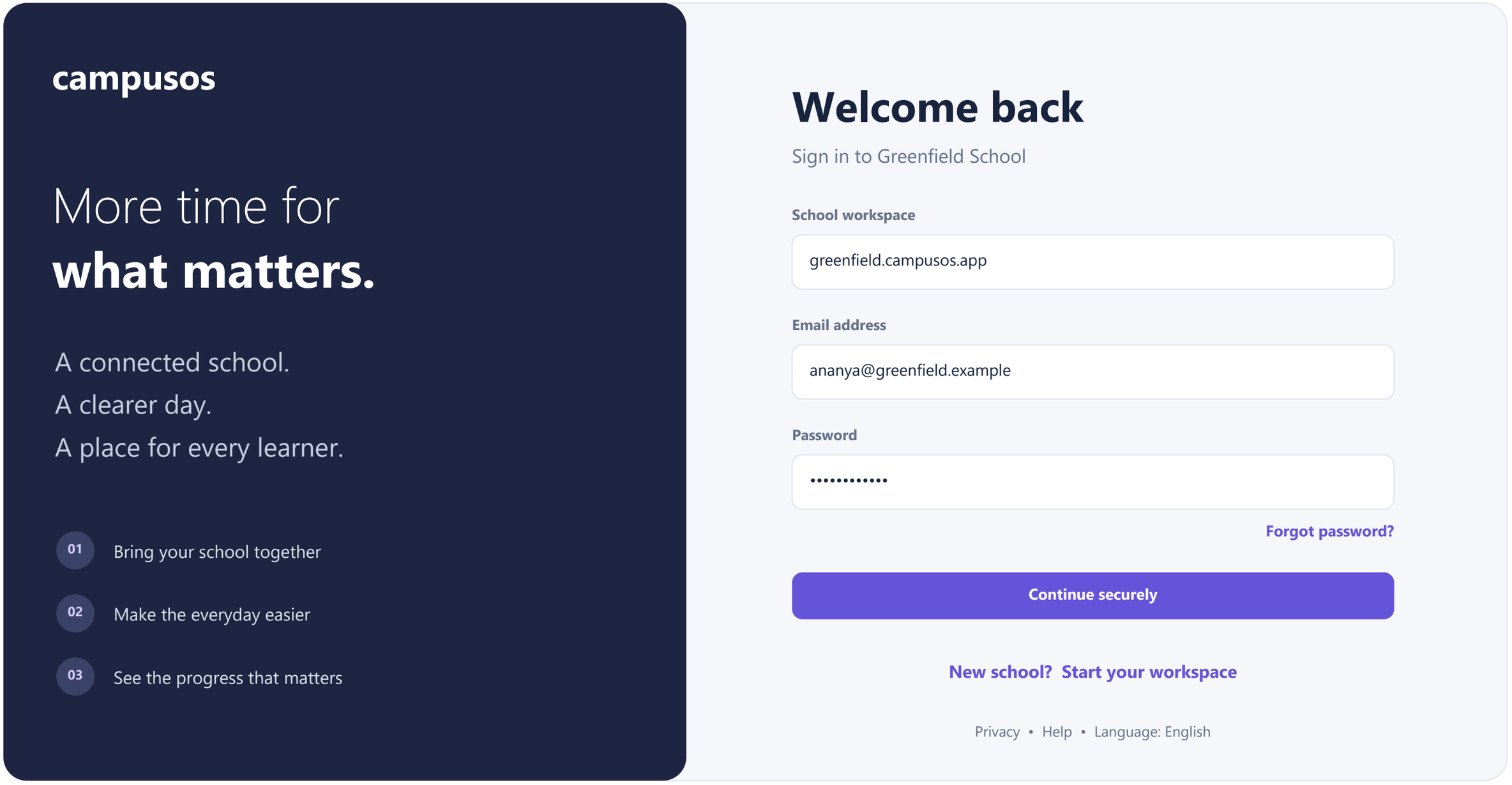
Interaction & accessibility [1]

Visible focus, keyboard actions, associated field labels and inline errors. Status uses text plus color; charts include tables. Respect reduced motion. Aim for 44 px touch controls, sufficient contrast and screen-reader announcements for async results.

Screens are high-fidelity static concepts with fictional data. They demonstrate layout and behavior intent; production accessibility must be verified.

S01 / Welcome to your school workspace

Identity and onboarding • TEN-01 to TEN-05, IAM-01 • desktop concept

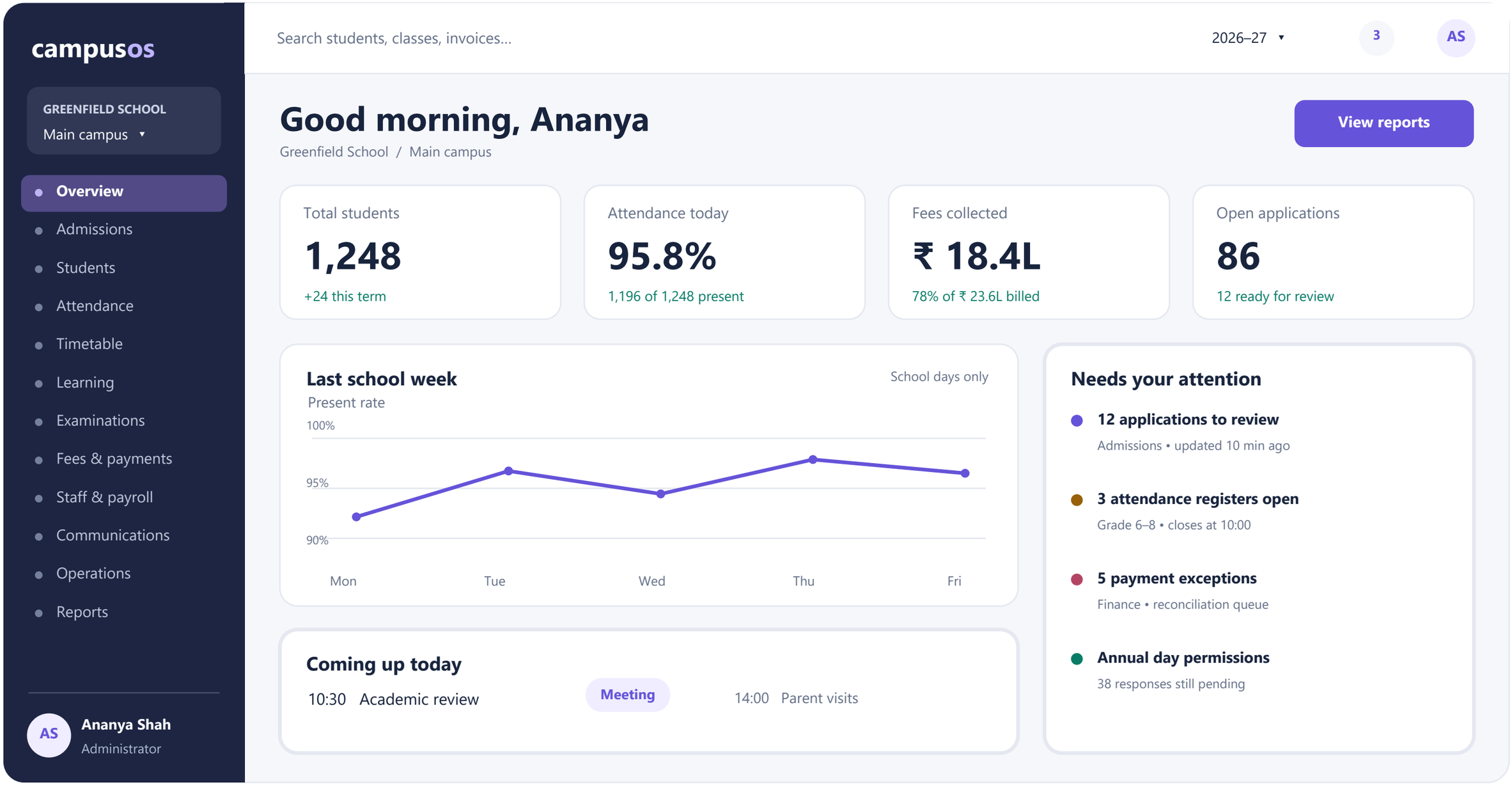


Primary flow: Verified sign-in → MFA when required → role workspace.

Recovery & access: Expired invitation and recovery states retain clear next steps.

S02 / Good morning, Ananya

School overview • TEN, ATT, FEE, ANA • one place to start the school day

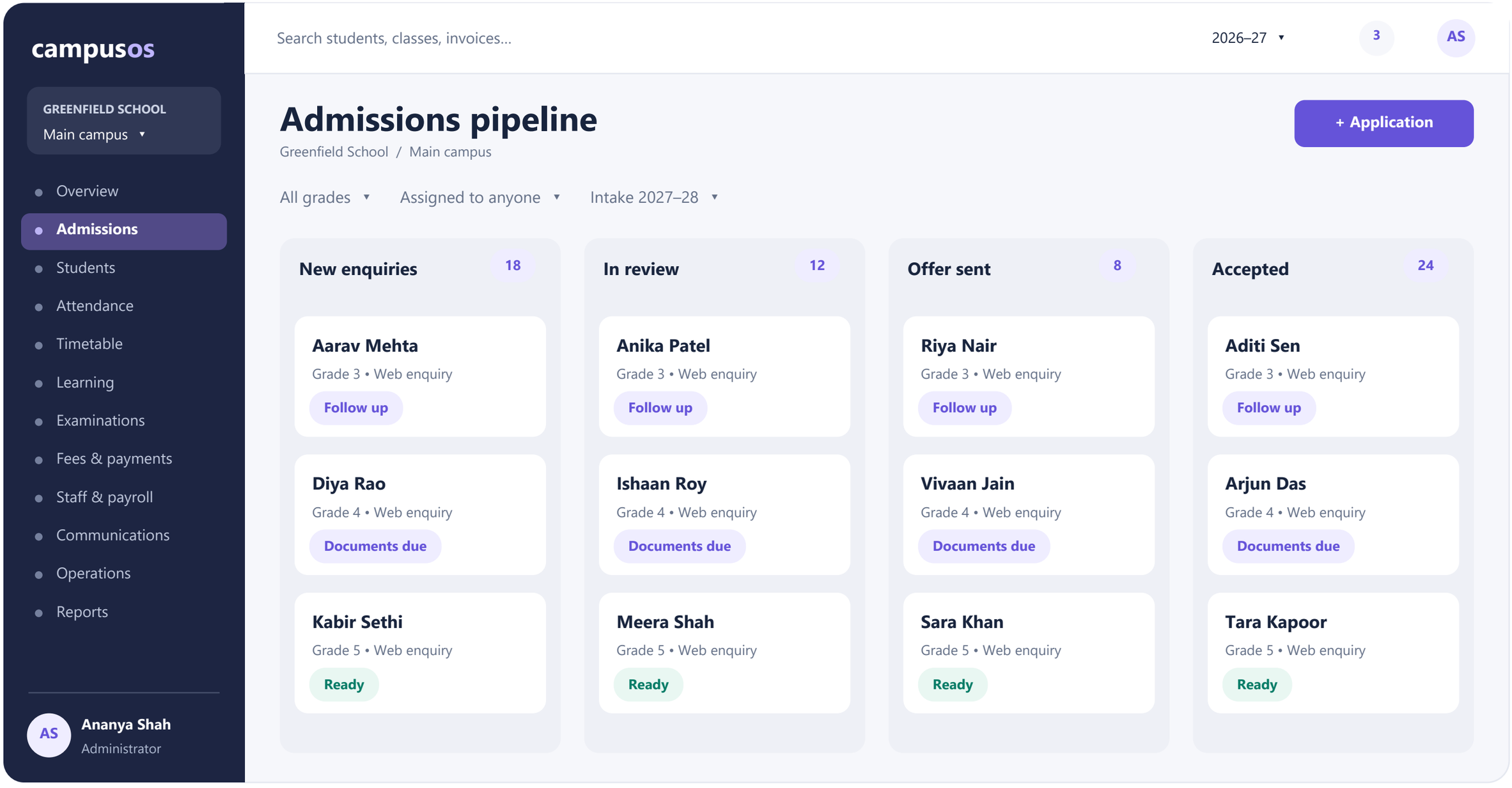


Primary flow: Open an exception, complete its action, return to the filtered overview.

Recovery & access: Every metric shows the applicable campus, date and authorized drilldown.

S03 / Admissions pipeline

Admissions • ADM-01 to ADM-06 • intake 2027–28



Primary flow: Review application → validate documents → offer → reserve seat → enrol.

Recovery & access: Capacity and offer expiry are checked before acceptance; changes are audited.

S04 / Student directory

Student records • SIS-01 to SIS-04 • searchable, scoped and easy to scan

campusos

GREENFIELD SCHOOL
Main campus ▾

- Overview
- Admissions
- **Students**
- Attendance
- Timetable
- Learning
- Examinations
- Fees & payments
- Staff & payroll
- Communications
- Operations
- Reports

AS **Ananya Shah**
Administrator

Search students, classes, invoices...

2026–27 ▾ 3 AS

Student directory

Greenfield School / Main campus

+ Add student

Active students	New this term	Documents due
1,248	**24**	**18**
Across 42 sections	Enrolment confirmed	Requires office follow-up

Search name or admission number... Grade: All ▾ Section: All ▾ Export ↓

STUDENT	ADMISSION NO.	GRADE / SECTION	GUARDIAN	STATUS
Aarav Mehta	GF-2026-0142	6 / A	Neha Mehta	Active
Anika Patel	GF-2026-0143	6 / A	Rohan Patel	Active
Diya Rao	GF-2025-0089	6 / B	Kavya Rao	Docs due
Ishaan Roy	GF-2024-0118	7 / A	Ritu Roy	Active
Meera Shah	GF-2025-0241	7 / B	Nisha Shah	Active

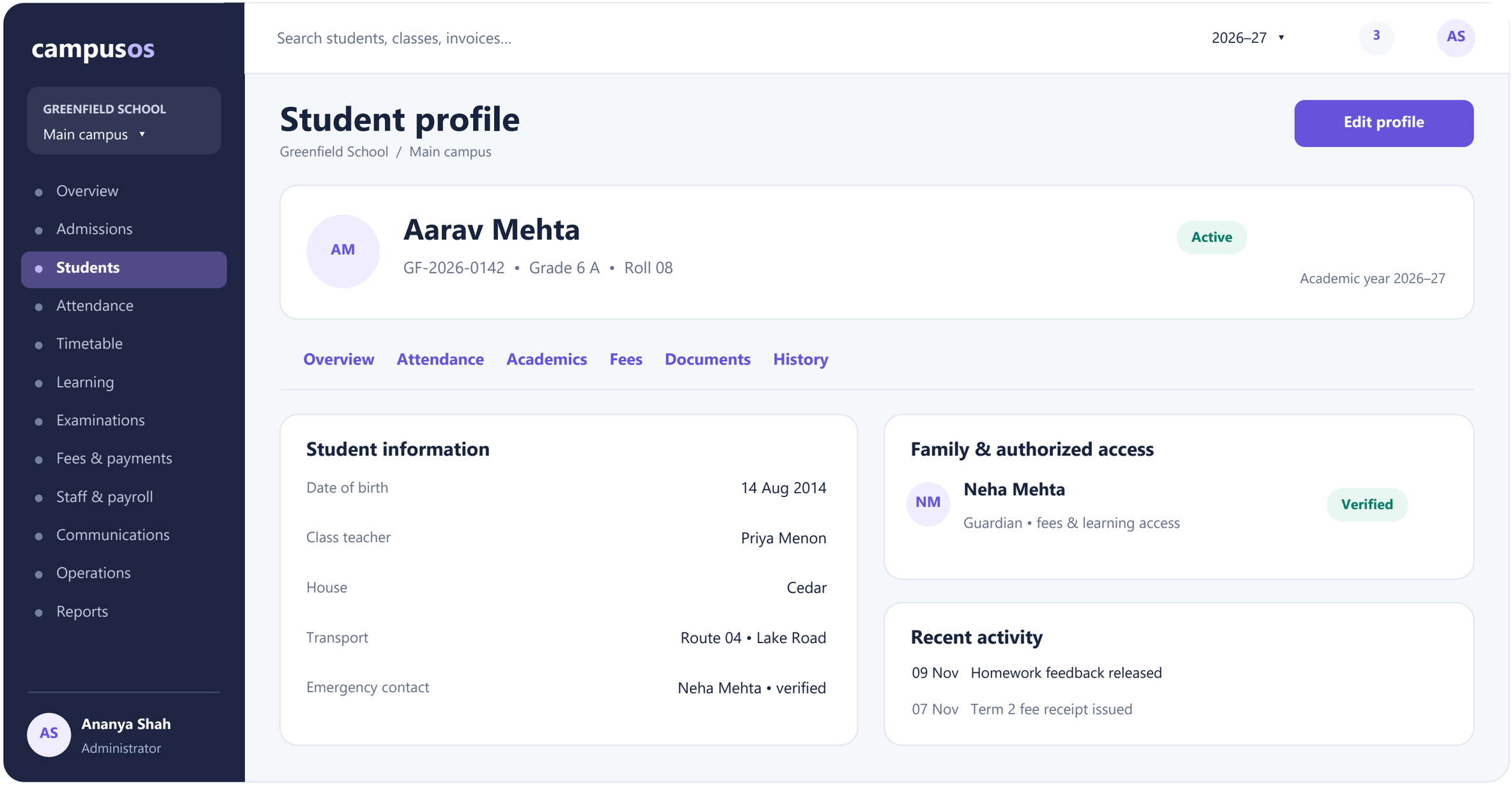
Showing 1–5 of 1,248 students

Primary flow: Search/filter → student record → approved update or certificate.

Recovery & access: Exports are asynchronous and scoped; duplicate records go through review.

S05 / Student profile

Student lifecycle • SIS, HLT, FAM • a connected record with restricted sensitive panels

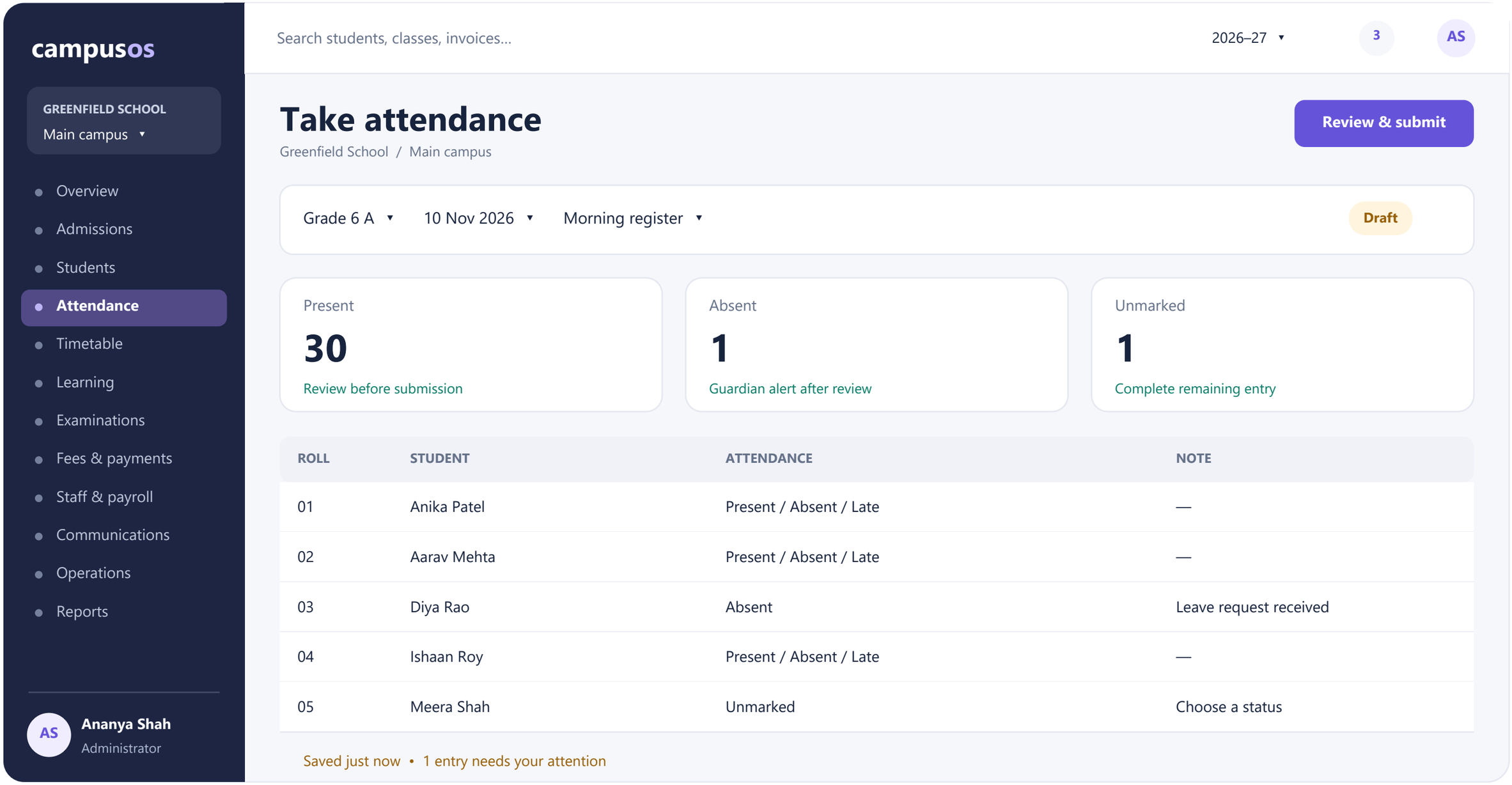


Primary flow: Review academic and family history before making a controlled change.

Recovery & access: Health/custody details appear only for explicit permissions; all reads are audited.

S06 / Take attendance

Class attendance • ATT-01 to ATT-06 • fast entry with a deliberate review step

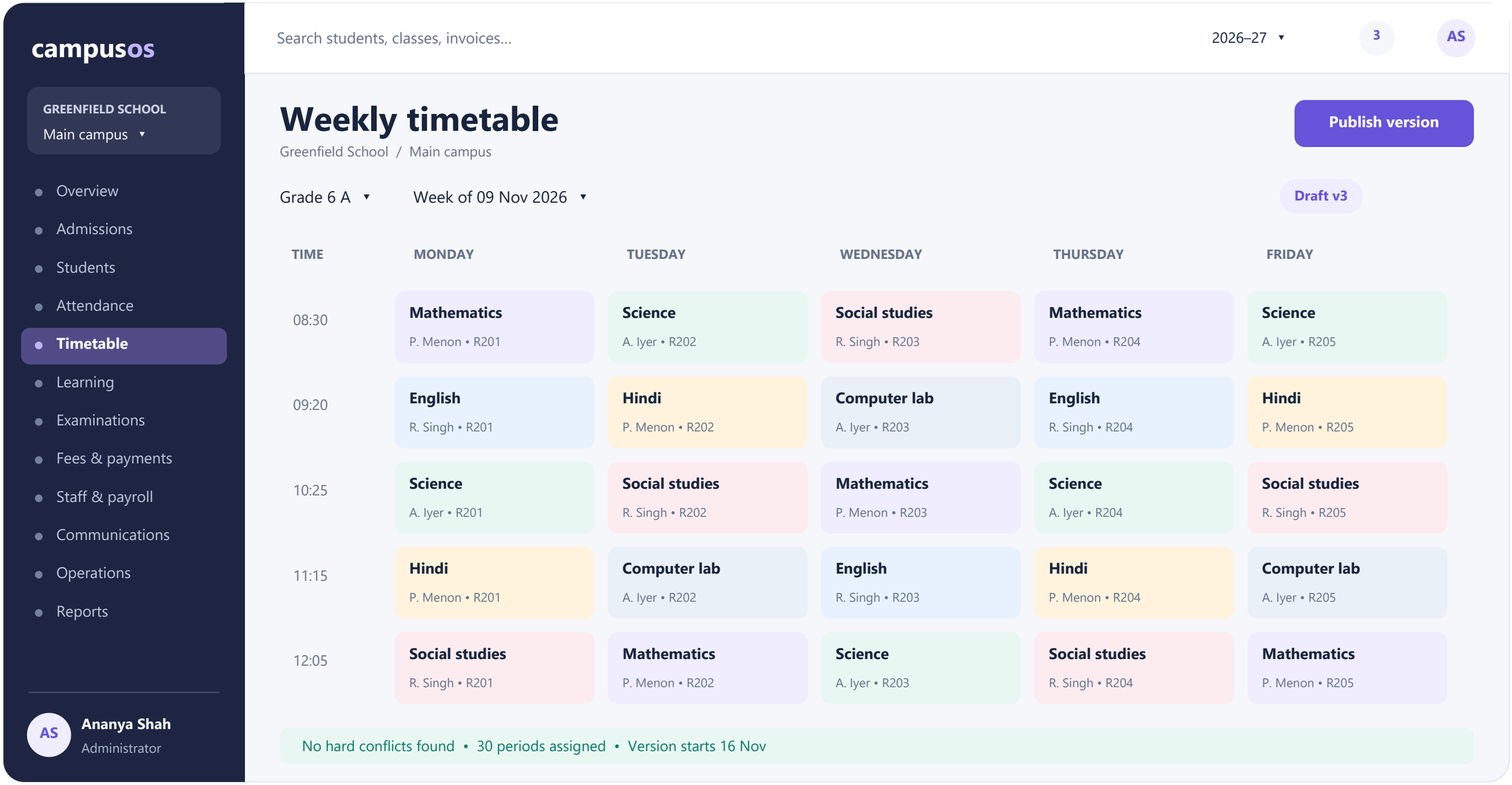


Primary flow: Mark each pupil → review unmarked entries → finalize → notify permitted guardians.

Recovery & access: Offline mode shows queued changes; conflicting edits cannot silently overwrite.

S07 / Weekly timetable

Academic scheduling • ACA, TIM • a clear weekly grid with conflict detection

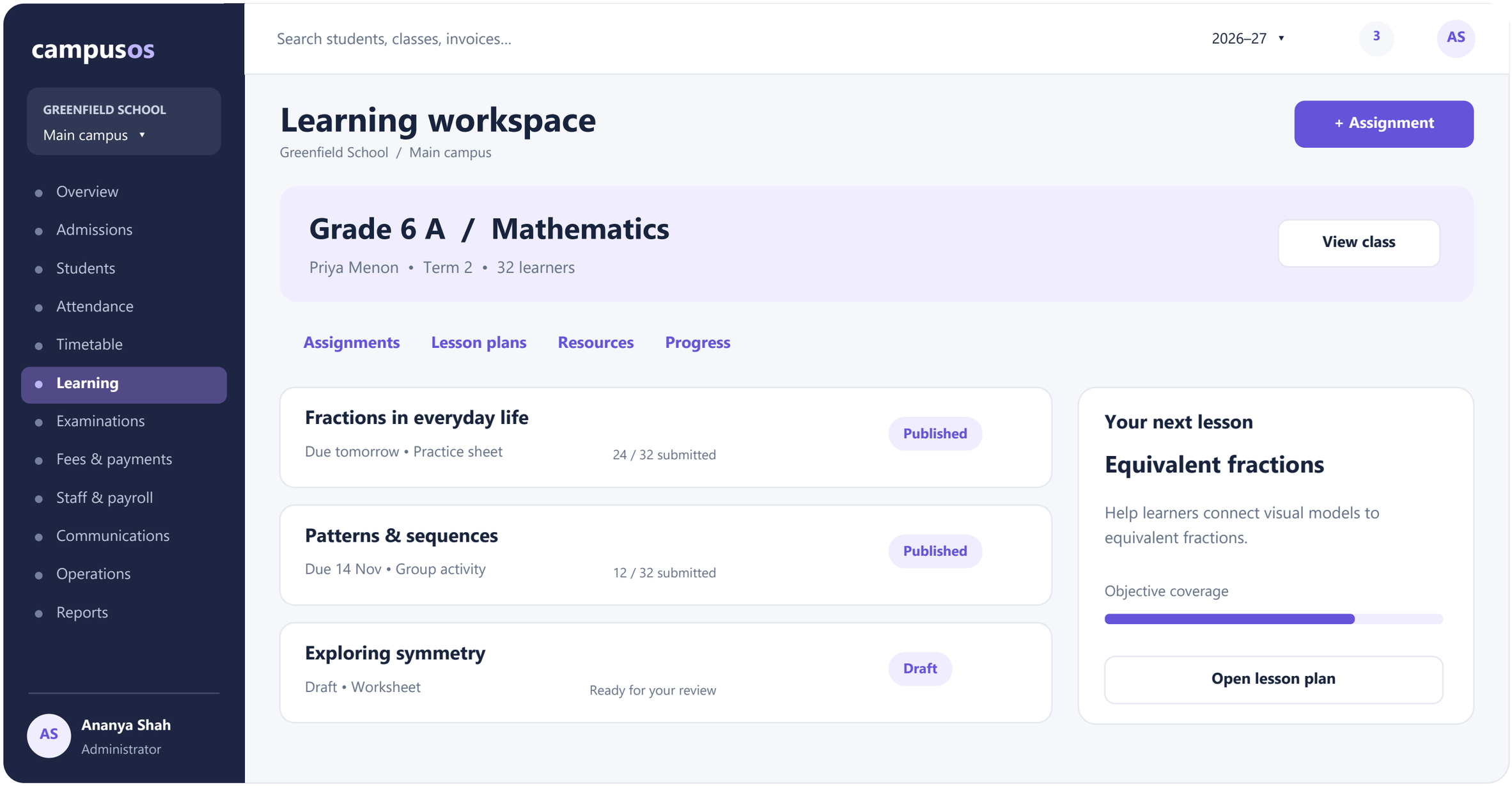


Primary flow: Edit draft → resolve teacher/room conflicts → review workload → publish.

Recovery & access: Hard conflicts block publication; an effective date preserves historical timetables.

S08 / Learning workspace

Teaching & assignments • LMS • keep preparation, work and feedback together



Primary flow: Prepare lesson → publish assignment → review submissions → release feedback.

Recovery & access: Draft scores stay private; individual extensions preserve original class deadlines.

S09 / Results & report cards

Assessment • EXM-01 to EXM-06 • review before publication

campusos

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Main campus ▾

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Administrator

Search students, classes, invoices...

2026–27 ▾ 3 AS

Results & report cards

Greenfield School / Main campus

Review results

Grade 6 / Term 2
128
Students in this cohort

Marks complete
96%
5 entries still required

Report status
Draft
Approval pending

STUDENT	ENGLISH / 100	MATHS / 100	SCIENCE / 100	REVIEW
Aarav Mehta	86	92	88	Ready
Anika Patel	91	89	94	Ready
Diya Rao	84	—	87	Missing
Ishaan Roy	78	85	82	Ready

Publication needs one more review

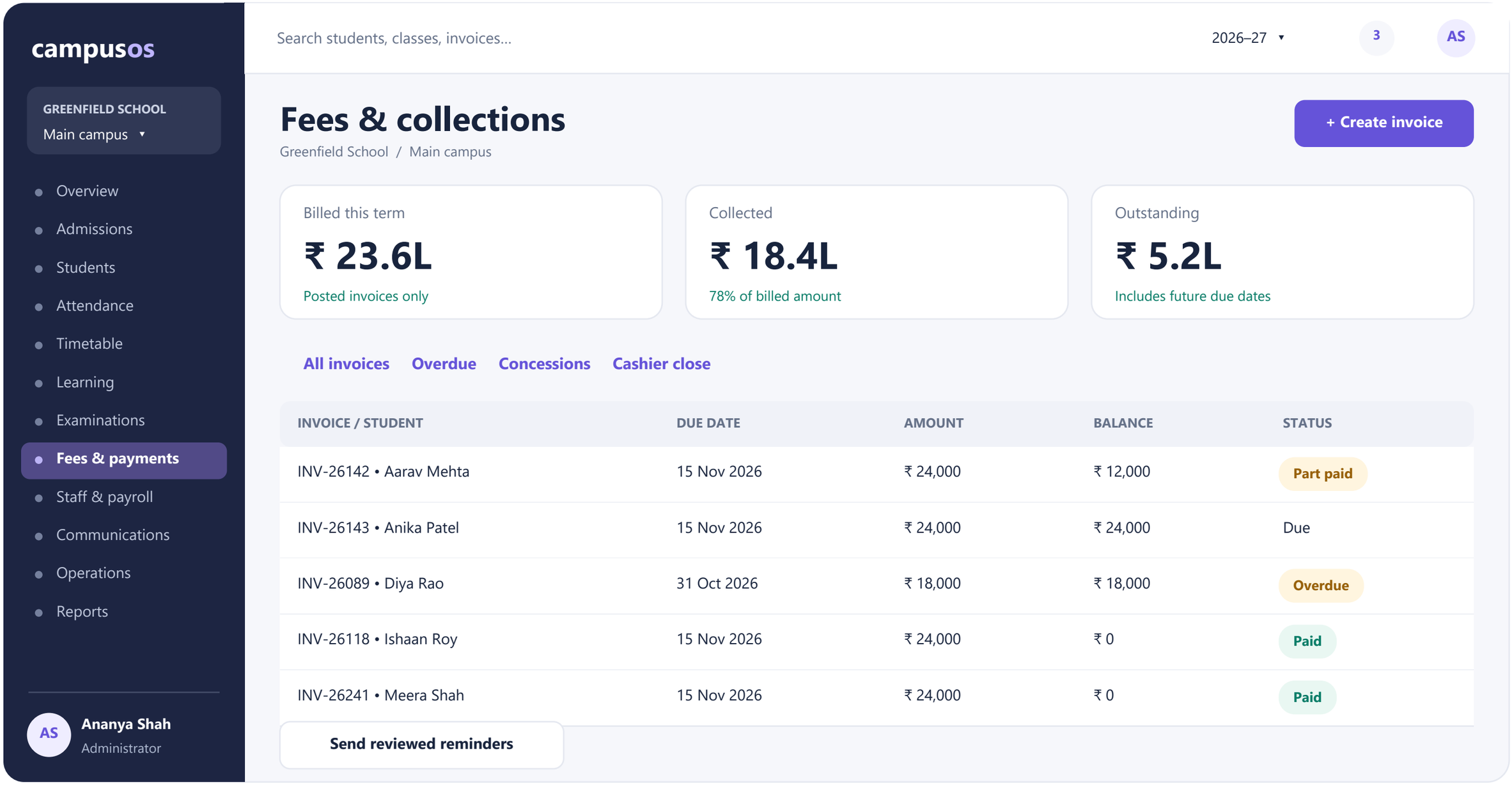
Resolve missing marks, verify the grading scheme and obtain coordinator approval.

Primary flow: Enter marks → validate → submit → moderate → approve → publish report snapshots.

Recovery & access: Missing required marks block release; corrections produce versioned reissues.

S10 / Fees & collections

Finance • FEE, FIN • explainable balances and controlled adjustments



Primary flow: Filter receivables → open ledger → collect, allocate or request adjustment.

Recovery & access: Posted entries use reversals; refunds/waivers require independent approval.

S11 / Payment reconciliation

Payment operations • PAY, FIN • compare capture, receipt and settlement

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Search students, classes, invoices...

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Payment reconciliation

Greenfield School / Main campus

Import statement

Matched today
142
Verified and allocated

Needs review
5
Assigned to finance

Settlement batch
₹ 2.84L
Net expected bank credit

Unmatched payments

REFERENCE	AMOUNT	REASON
pay_demo_102	₹ 12,000	Receipt pending
pay_demo_108	₹ 8,500	Invoice mismatch
pay_demo_114	₹ 24,000	Settlement pending
pay_demo_119	₹ 6,000	Review required

Review • pay_demo_102

Student	Aarav Mehta
Invoice	INV-26142
Provider status	Captured
Verified amount	₹ 12,000
Receipt status	Not posted

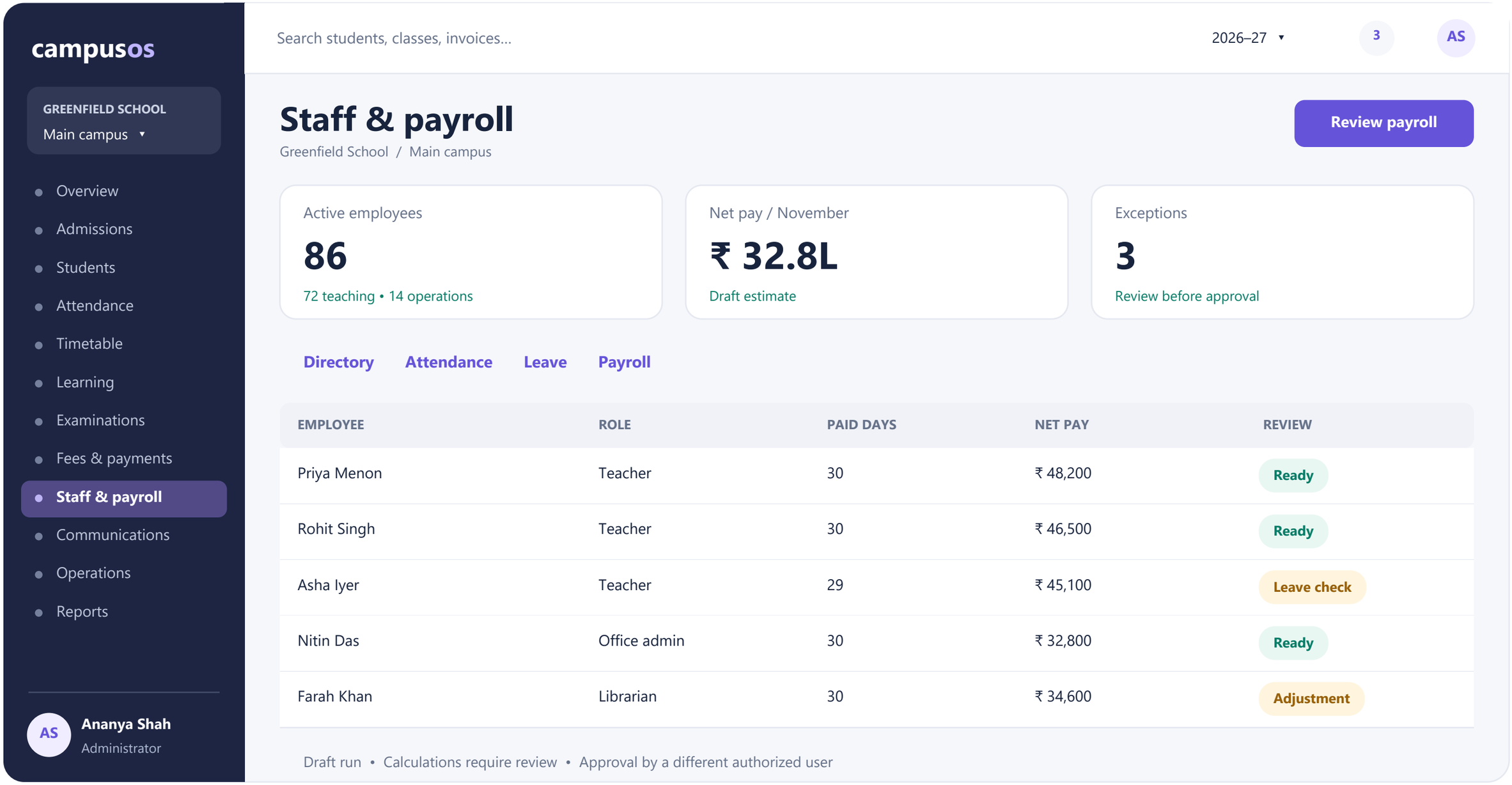
Retry verified posting

Primary flow: Inspect verified evidence → resolve exception → post safely → reconcile bank.

Recovery & access: Repeated events or replay cannot create a second receipt or allocation.

S12 / Staff & payroll

HR and payroll • HRM • a reviewable monthly run with restricted access

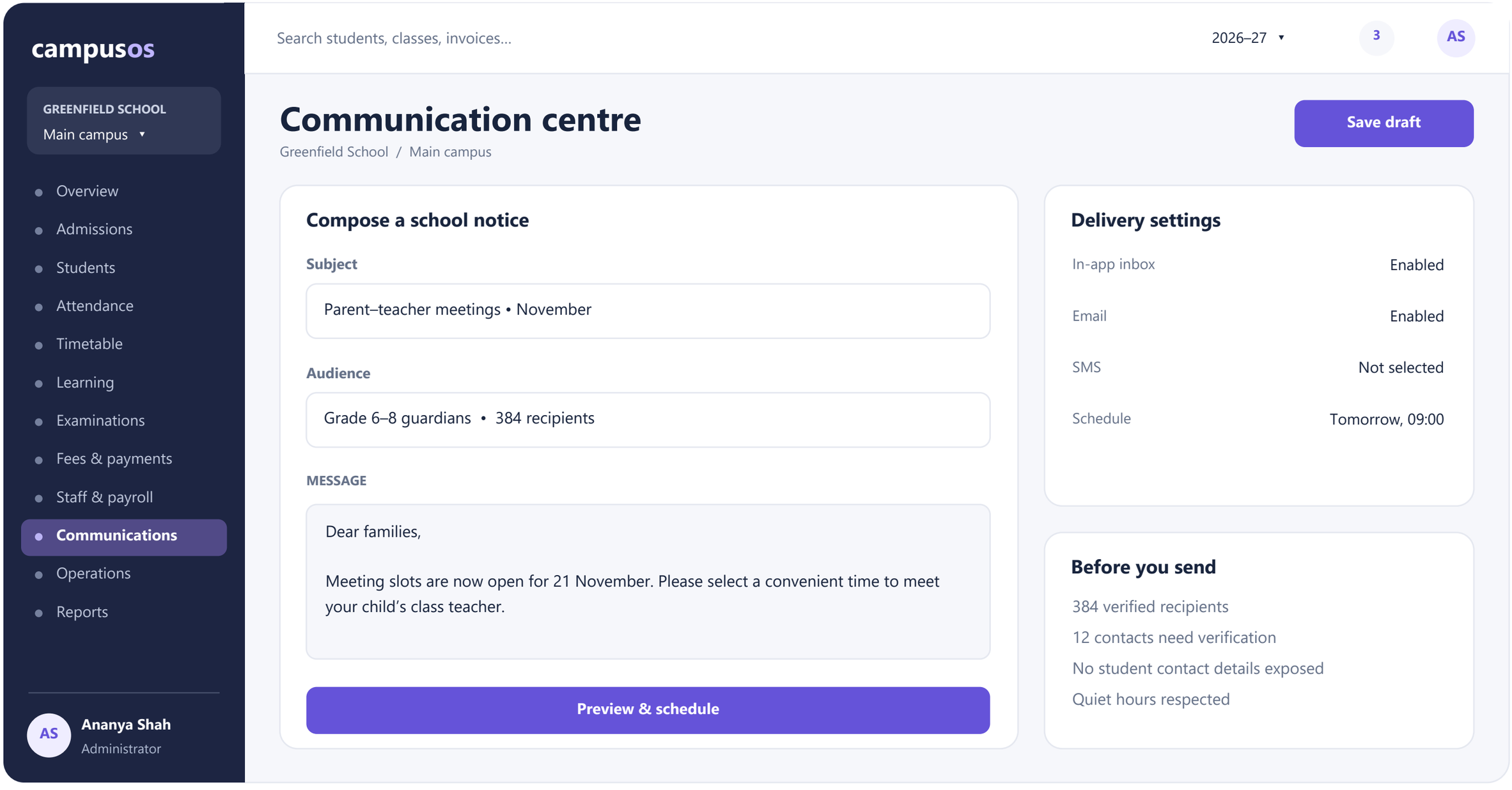


Primary flow: Reconcile inputs → draft payroll → resolve exceptions → independent approval.

Recovery & access: Salary and bank fields require HR scope; approved corrections remain versioned.

S13 / Communication centre

Notices, meetings and messages • COM • audience clarity before sending



Primary flow: Select audience → preview recipients/content → approve → schedule → monitor.

Recovery & access: Consent, verified contacts, restricted guardians and quotas filter the audience.

S14 / Transport operations

Routes and live trips • TRN • route diagram is illustrative, not a real map

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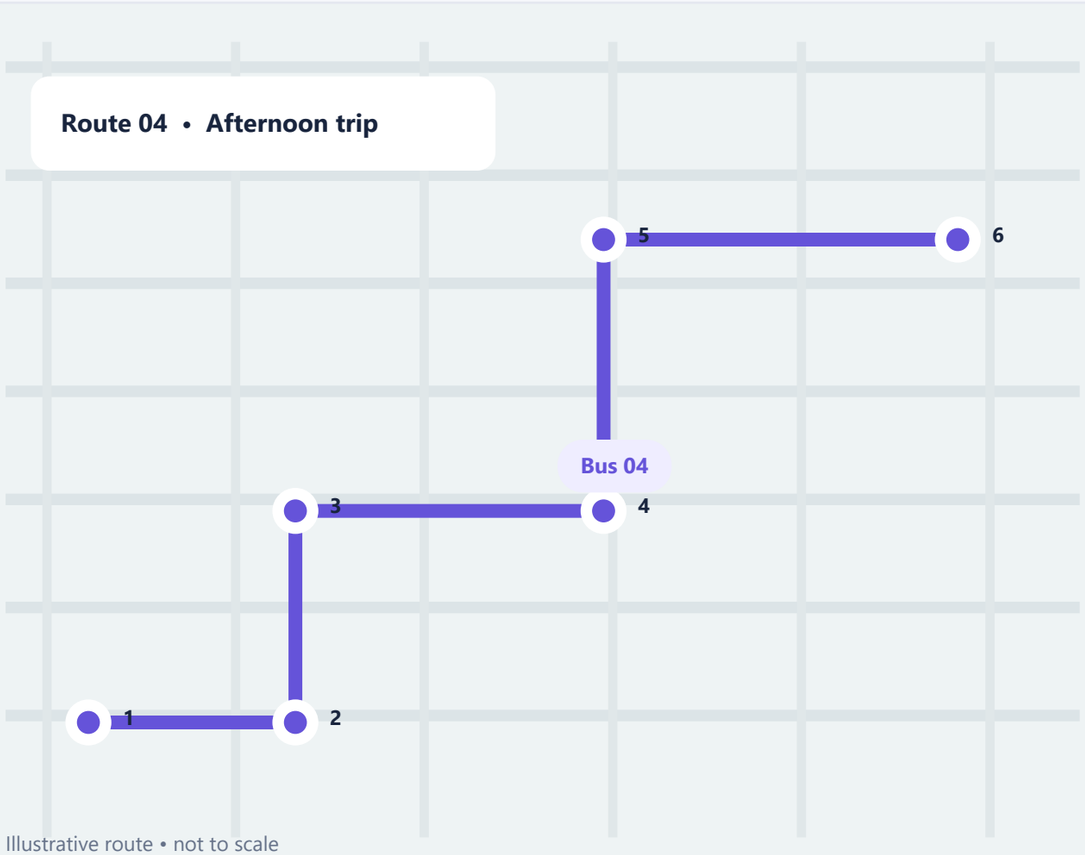
Search students, classes, invoices…

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Transport operations

Greenfield School / Main campus

Manage routes



Illustrative route • not to scale

Route 04 • Lake Road

On schedule

Last update: 12 seconds ago

On board	28 / 32
Driver	R. Kumar
Attendant	S. Devi
Next stop	Cedar Avenue

Trip activity

15:12 Departed school

15:18 Stop 1 completed

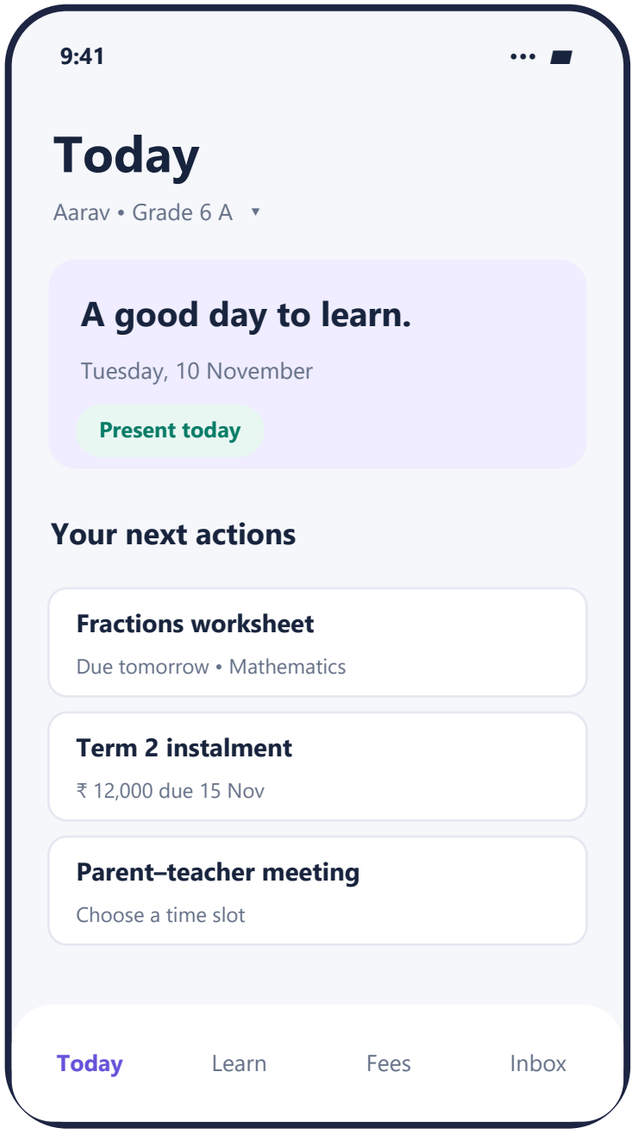
15:23 Stop 2 completed

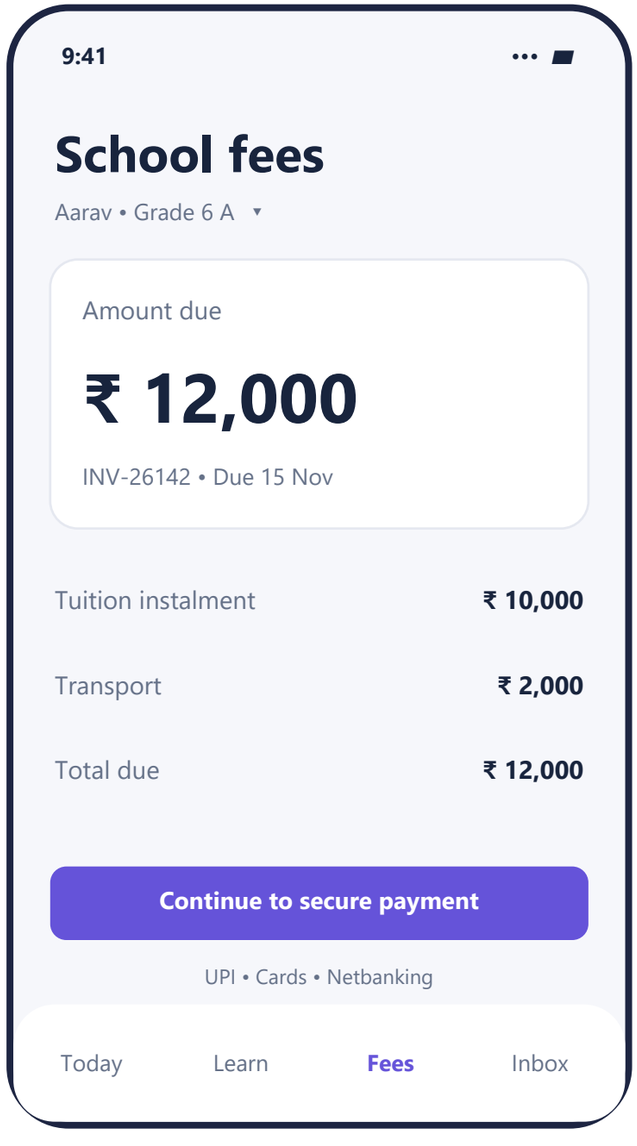
Primary flow: Review assigned route → board/alight roster → record exceptions and handover.

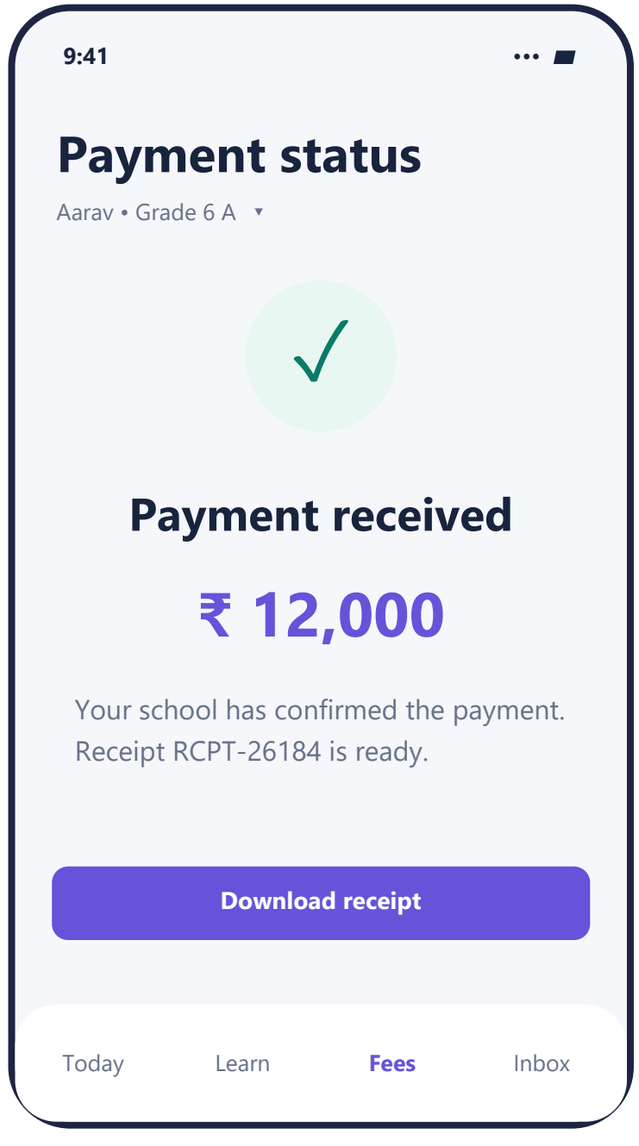
Recovery & access: Stale location displays its age and no ETA; families see only authorized trips.

S15 / The family day, beautifully simple.

Parent portal • FAM, FEE, PAY • three connected mobile views





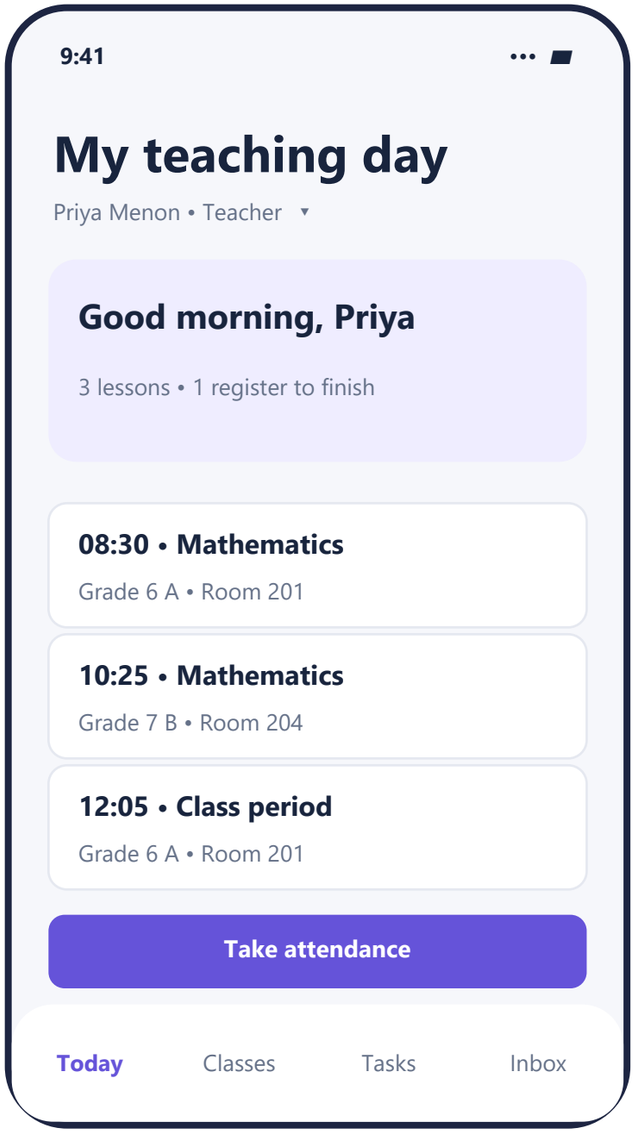


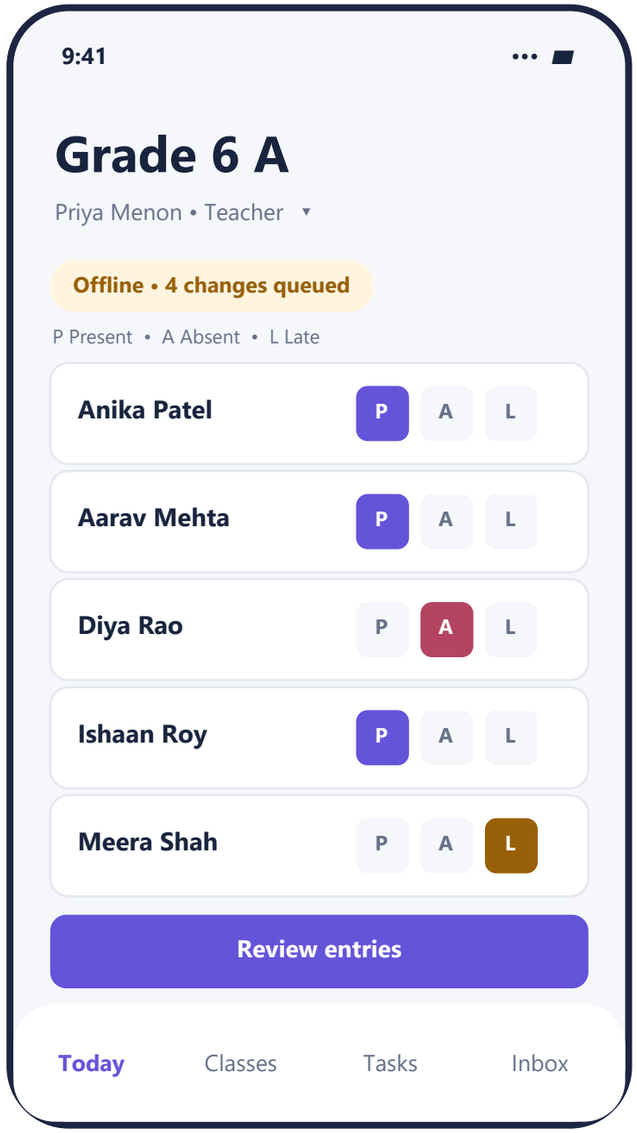
Primary flow: Home → invoice detail → hosted payment → verified receipt.

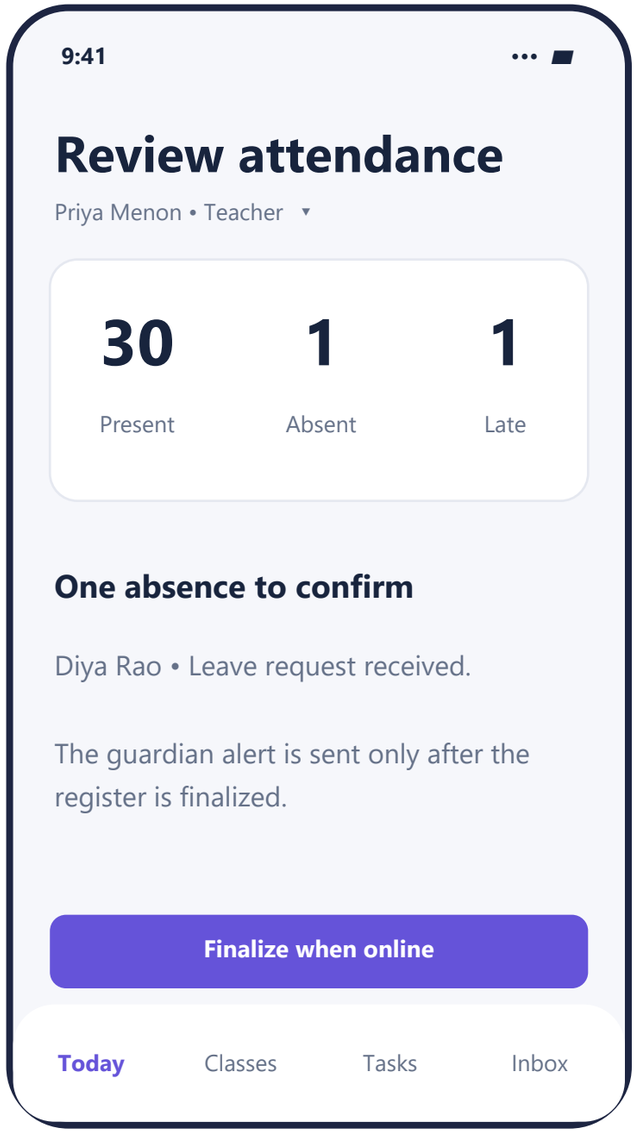
Recovery & access: If confirmation is delayed, show pending and discourage duplicate payment.

S16 / Attendance from the classroom.

Teacher mobile • ATT • touch-first entry with visible offline and review states





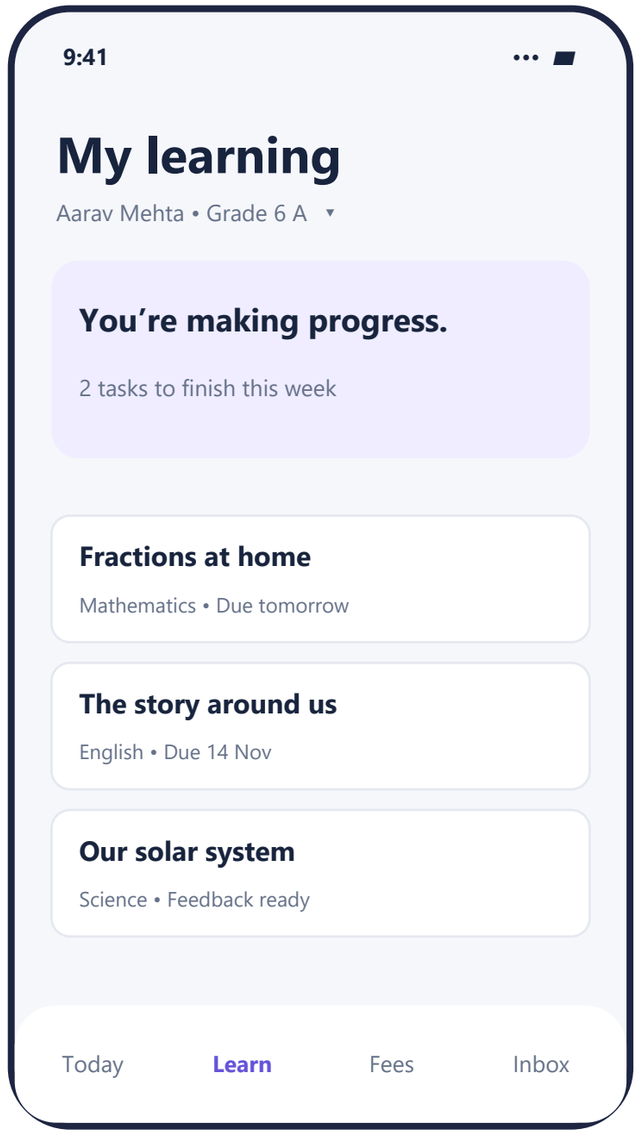


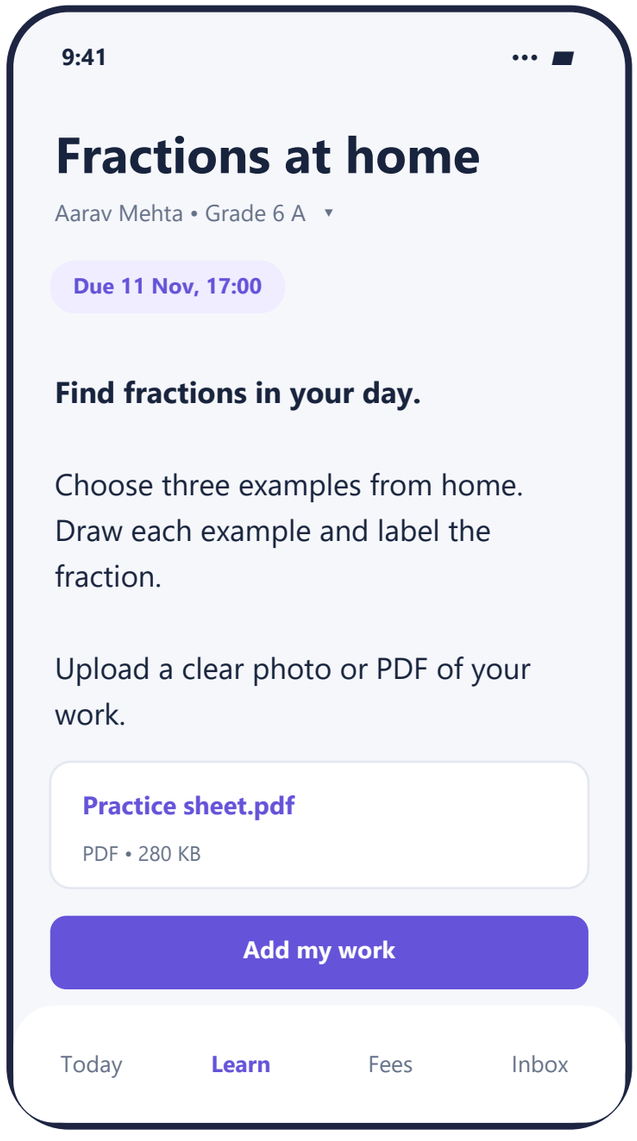
Primary flow: Open class → mark students → review totals → finalize when synchronized.

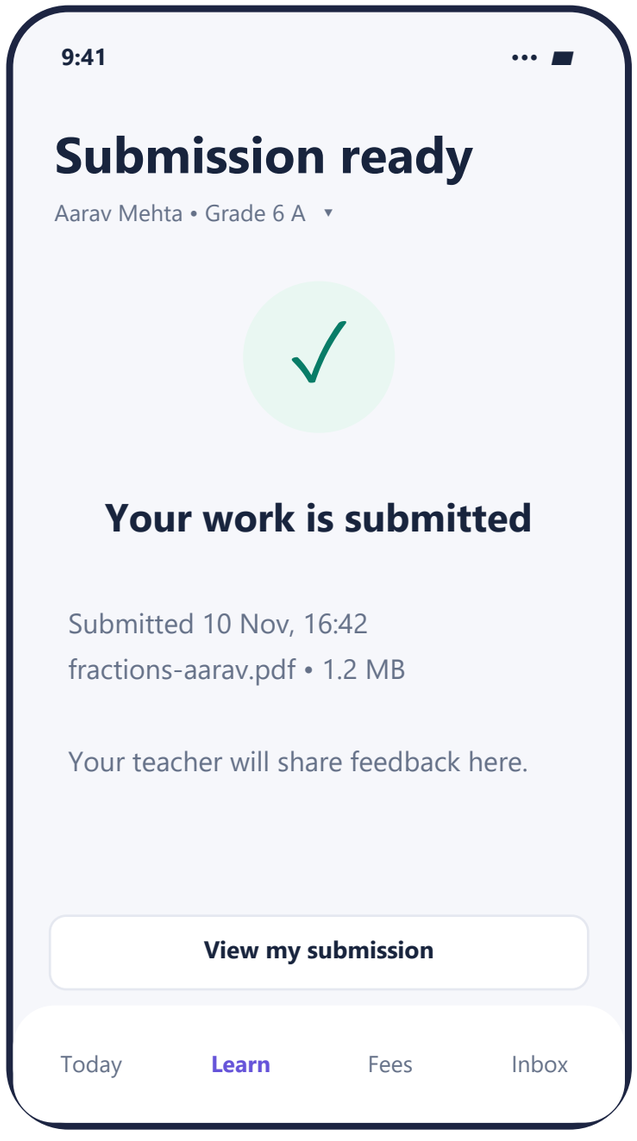
Recovery & access: Offline changes are drafts; conflicting newer corrections need explicit resolution.

S17 / Learning with a clear next step.

Student mobile • LMS, FAM • from assignment to a confirmed submission







Primary flow: Find assigned work → read instructions → upload → confirm submission.

Recovery & access: Uploads show progress/retry; allowed resubmission creates a new version.

S18 / Roles & access

Administration • IAM, INT • explicit permissions with understandable scopes

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Search students, classes, invoices...

2026–27 ▾ 3 AS

Roles & access

Greenfield School / Main campus

Save role

Role: Class teacher ▾ **Template: Teaching staff**

Role templates

- School owner
- Principal
- **Class teacher**
- Finance officer
- School nurse
- Librarian
- Guardian

CAPABILITY	ACTION	SCOPE
Student basics	View	Assigned classes
Attendance	Create / correct	Assigned sessions
Homework	Manage	Assigned subjects
Marks	Draft / submit	Assigned subjects
Final results	View published	Assigned classes
Financial records	No access	—
Health records	Care alert only	Assigned pupils

Primary flow: Start from template → adjust bounded actions/scopes → preview access → save.

Recovery & access: No role grants beyond the editor's authority; sensitive changes require audit.

S19 / SaaS operations

Platform control plane • PLT • operational metadata without child records

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PLATFORM
Operator workspace

- Overview
- Tenants
- Subscriptions
- Usage & billing
- Onboarding
- Service health
- Support
- Audit log

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Search students, classes, invoices...

2026–27 ▾ 3 AS

SaaS operations

CampusOS / Service operations

+ Create tenant

Active schools
124
8 onboarding this month

Active students
1.82L
Illustrative billable usage

Service availability
99.97%
Illustrative last 30 days

SCHOOL	PLAN	STUDENTS	LIFECYCLE	HEALTH
Greenfield School	Growth	1,248	Active	Healthy
Cedar Valley Academy	Core	842	Active	Healthy
Northstar Public School	Growth	2,180	Onboarding	Setup 72%
Maple International	Enterprise	4,920	Active	Healthy
Hillview School	Core	615	Grace period	Billing review

Support access requires an approved school grant. All service metrics shown here are fictional.

Primary flow: Review tenant lifecycle/health → resolve onboarding, usage or billing tasks.

Recovery & access: School data requires a separate time-limited grant; suspension follows policy.

S20 / Library desk

Resource circulation • LIB • books, borrowers and availability in one view

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Search students, classes, invoices...

2026–27 ▾ 3 AS

Library desk

Greenfield School / Main campus

Issue a book

Copies available	On loan	Overdue
4,216	384	18
Across 1,780 titles	Across 296 borrowers	Reminders ready for review

Scan barcode or search title, author, accession number...

FICTION — Available
The Blue Umbrella
Ruskin Bond
View copies

MATHEMATICS — Available
A World of Numbers
Learning collection
View copies

SCIENCE — Available
Our Living Planet
Science collection
View copies

NON-FICTION — Available
Stories of Discovery
Reading collection
View copies

Primary flow: Scan borrower + copy → validate limits → issue → renew or return.

Recovery & access: Only one open loan per copy; lost/damaged fines use reviewed policies.

S21 / Purchasing & inventory

Procurement • INV • clear status from request to receipt

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Search students, classes, invoices...

2026–27 ▾ 3 AS

Purchasing & inventory

Greenfield School / Main campus

+ Requisition

Awaiting approval	Open orders	Low-stock items
8	14	6
3 above department limit	₹ 4.2L committed	Review reorder quantities

REQUEST	DEPARTMENT	VALUE	STAGE	OWNER
REQ-041 • Lab supplies	Science	₹ 48,000	Approval	A. Iyer
REQ-042 • Library books	Library	₹ 32,500	Ordered	F. Khan
REQ-043 • Sports kits	Sports	₹ 18,200	Receiving	R. Singh
REQ-044 • Projector	Academics	₹ 42,000	Approval	P. Menon
REQ-045 • Stationery	Office	₹ 8,400	Received	N. Das

Requester → Budget check → Approval → Purchase order → Goods receipt → Reconciliation

Primary flow: Review request and budget → approve → order → receive quantities → reconcile.

Recovery & access: Partial receipts keep orders open; requester cannot approve controlled spend.

S22 / Help & service desk

Support and operations • OPS, PLT • shared accountability with scoped conversations

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Search students, classes, invoices...

2026–27 ▾ 3 AS

Help & service desk

Greenfield School / Main campus

+ New ticket

Your queues

Open tickets	12
Awaiting school	4
Assigned to me	5
Resolved this week	28

Need product support?

Create a time-limited support grant only when needed.

TKT-1042 • Payment receipt not visible

In progress

Owner: Finance team • Priority: Normal

Neha Mehta • Parent

My payment shows as pending. Please help me confirm the receipt for invoice INV-26142.

Finance team • Visible to parent

We are checking the provider confirmation. Please wait before making another payment.

Reply to parent Internal note / Add attachment

Primary flow: Classify → assign → investigate → reply → resolve with an auditable outcome.

Recovery & access: Internal notes stay private; platform support requires a separate school grant.

S23 / School insights

Analytics • ANA • explainable trends and permission-aware drilldowns

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- **Reports**

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Search students, classes, invoices…

2026–27 ▾ 3 AS

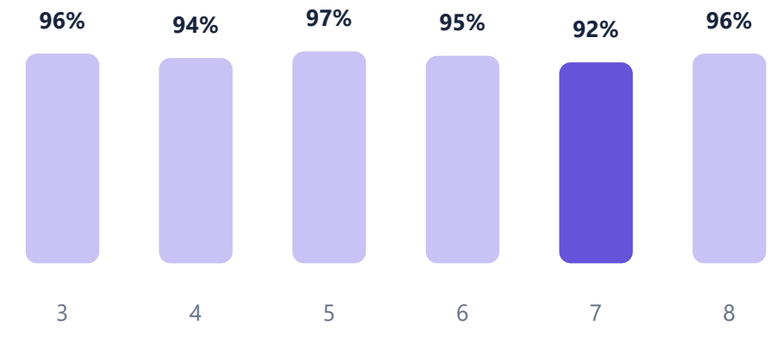
School insights

Greenfield School / Main campus

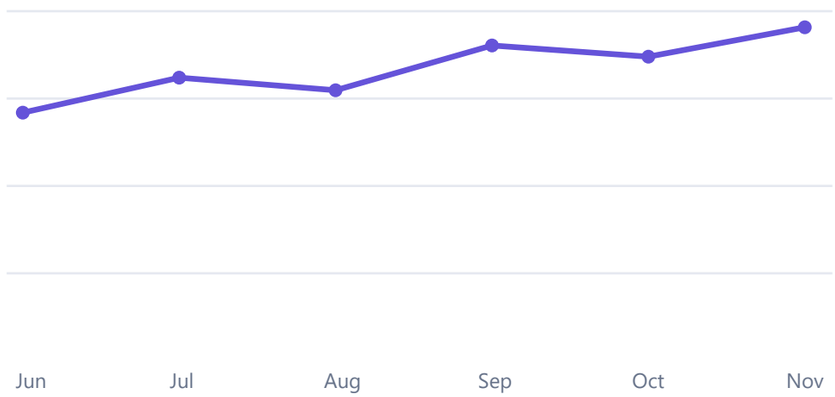
Export report

Academic year 2026–27 ▾ Main campus ▾ Term 2 ▾

Attendance by grade



Monthly fee collections



Metric definitions are part of the report

Attendance excludes holidays and unmarked entries. Collections reflect verified receipts, not bank settlement.

Primary flow: Choose scope → inspect definitions → drill into records → schedule/export.

Recovery & access: Filters apply to charts and downloads; small sensitive cohorts are restricted.

Design the difficult moments, too.

These states are part of every applicable screen’s definition of done.

EMPTY / FIRST USE

No students yet

Import your student list or add your first student.

Import students

VALIDATION / INLINE

A little more information

Add the required due date before creating this invoice.

Review highlighted fields

OFFLINE / QUEUED

Saved on this device

4 attendance changes are waiting to sync. Keep this device secure.

Retry connection

CONFLICT / STALE DATA

This register has changed

Another teacher saved a newer version. Review both changes.

Compare changes

PAYMENT / PENDING

We’re confirming your payment

You can safely leave this page. Please wait before paying again.

Check payment status

ACCESS / PERMISSION

This record is restricted

Your current role does not include this action. Contact your school admin.

Return to workspace

Also implement loading skeletons, no search results, expired sessions, failed uploads, safe destructive confirmation and success feedback.

Screen index & requirement traceability.

The 23 screen concepts cover representative high-value journeys. Related modules reuse the same component system.

ID	SCREEN / EXPERIENCE	REQUIREMENTS	PAGE
S01	Welcome to your school workspace	TEN / IAM	46
S02	Good morning, Ananya	TEN / ATT / FEE / ANA	47
S03	Admissions pipeline	ADM	48
S04	Student directory	SIS	49
S05	Student profile	SIS / HLT / FAM	50
S06	Take attendance	ATT	51
S07	Weekly timetable	ACA / TIM	52
S08	Learning workspace	LMS	53
S09	Results & report cards	EXM	54
S10	Fees & collections	FEE / FIN	55
S11	Payment reconciliation	PAY / FIN	56
S12	Staff & payroll	HRM	57
S13	Communication centre	COM	58
S14	Transport operations	TRN	59
S15	The family day, beautifully simple.	FAM / FEE / PAY	60
S16	Attendance from the classroom.	ATT	61
S17	Learning with a clear next step.	LMS / FAM	62
S18	Roles & access	IAM / INT	63
S19	SaaS operations	PLT	64
S20	Library desk	LIB	65
S21	Purchasing & inventory	INV	66
S22	Help & service desk	OPS / PLT	67
S23	School insights	ANA	68

Each module also includes an acceptance scenario. The companion CSV lists every functional requirement with module, phase and screen references.

Source notes & implementation handoff.

Primary references checked for this blueprint on 10 November 2026. Product requirements are original recommendations.

[1]

Accessibility • W3C: Web Content Accessibility Guidelines 2.2

Target WCAG 2.2 AA; visual mockups alone do not establish conformance.

Open official source ↗

[2]

Security • OWASP: Application Security Verification Standard

Use ASVS 5.0 as a verification catalogue. This blueprint does not constitute certification.

Open official source ↗

[3]

India data protection • MeitY: DPDP Rules 2025, corrigendum and enforcement timeline

Validate current applicability, commencement and child-data obligations before launch; official material includes phased enforcement.

Open official source ↗

[4]

Payment events • Razorpay: Validate and Test Webhooks

Verify signatures and handle duplicate/out-of-order events. Confirm current chosen-provider contracts.

Open official source ↗

[5]

Database isolation • PostgreSQL: Row Security Policies

Row policies add isolation; owners and privileged roles require special care. Test the deployed version.

Open official source ↗

Handoff: PDF + requirements CSV + screen previews + editable Python source + build/verification notes.